



Kolmogorov–Arnold networks-based GRU and LSTM for loan default early prediction

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HIGHLIGHTS

- KAN enhances the predictive power of GRU and LSTM for time series data.
- GRU-KAN and LSTM-KAN show resilience to concept drift, excelling with fewer training samples.
- The “blank interval” concept enables the prediction of default events three months early with 92% accuracy.
- The proposed models outperform baselines in stability across feature windows.

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ABSTRACT

This study addresses a key challenge in time series anomaly detection: the prediction of loan defaults several months in advance. Using real-world monthly repayment data, the study aims to provide financial institutions with a reliable basis for proactive intervention. Existing studies often neglect early prediction capabilities and rely on out-of-sample rather than out-of-time testing, limiting the evaluation of model adaptability to evolving financial conditions and constraining the effective use of historical data. To bridge this gap, this study introduces Kolmogorov–Arnold Network (KAN)-enhanced Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) models, denoted as GRU-KAN and LSTM-KAN, along with a “blank interval” concept to simulate early prediction scenarios. Evaluations compare these models against baseline methods, including LSTM, GRU, LSTM-Attention, and LSTM-Transformer, across varied resampling strategies, feature window lengths, blank interval durations, sample sizes, and multi-year and cross-quarter test sets. Ablation studies isolate the individual contributions of each model component. The results of the experiments indicate that the proposed models consistently outperform the baselines, with random undersampling and feature windows of 12 to 18 months producing better performance. GRU-KAN achieves optimal results for blank intervals of 1 to 8 months, while LSTM-KAN excels for intervals of 9 to 12 months.

1. Introduction

Time series anomaly detection focuses on identifying unusual patterns in sequential data. Such anomalies often indicate significant deviations from normal behavior and may signal potential risks. A key application in finance is loan default prediction, which plays a crucial role in risk management. Accurate prediction of loan defaults is essential for reducing non-performing loans and protecting asset quality [1]. Loan default occurs when a borrower fails to repay the principal or interest on

time or when the loan becomes extensively overdue. Predicting defaults requires analyzing extensive historical data and borrower characteristics using data analysis and machine learning techniques. Effective early prediction enables financial institutions to identify high-risk borrowers before default occurs and take preventive measures, thereby mitigating non-performing loans and enhancing overall financial health.

Although existing loan default prediction models such as LSTM can partially identify high-risk borrowers, they remain limited in predictive

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Nomenclature

Abbreviations

AUC	area under the ROC curve
BH-FDR	Benjamini–Hochberg false discovery rate
BY-FDR	Benjamini–Yekutieli false discovery rate
CLDS	current loan delinquency status
CNN	convolutional neural network

GRU	gated recurrent unit
KAN	Kolmogorov-Arnold network
LSTM	long short-term memory
MLP	multi-layer perceptron
OOS	out-of-sample
OOT	out-of-time
RNN	recurrent neural network
UPB	unpaid principal balance

accuracy and practical utility. In particular, predicting defaults several months in advance is crucial for financial institutions to minimize potential losses. However, existing research has paid insufficient attention to the early prediction capability of time series anomaly detection models, depriving financial institutions of the time needed to implement effective risk reduction strategies.

Existing models typically rely on training and testing with out-of-sample (OOS) data, meaning that both sets originate from the same year. This approach fails to evaluate the model's adaptability to changing financial environments, resulting in predictions that do not reflect current market conditions. A key challenge for researchers is the need to wait for sufficient data accumulation, which undermines the model's real-time applicability. This limitation not only prolongs model development but also restricts the effective use of extensive historical data for model training.

The novelty of this study lies in two key aspects. First, this study proposes two hybrid architectures, GRU-KAN and LSTM-KAN, designed to improve several-month-ahead loan default prediction. This modeling design addresses the limitations of existing anomaly detection models in accurately predicting loan defaults well before they materialize, thereby providing financial institutions with sufficient time for proactive risk management. By leveraging the nonlinear expressive capability of KAN to capture subtle and high-order feature interactions that remain informative even under several-month-ahead prediction settings, together with the temporal modeling strengths of recurrent neural networks, the proposed architectures are able to extract early-stage default signals more effectively than conventional models, and consequently demonstrate significantly superior performance when trained and evaluated on out-of-time (OOT) data across different years.

Second, this study proposes the concept of a “blank interval”, defined as a deliberately excluded temporal gap between the input feature window and the observation period. The blank interval is introduced to simulate the scenario of predicting several months in advance and to reflect the operational constraints of early-warning credit risk management. This design ensures that the model bases its predictions solely on information available well in advance, rather than on the most recent months that may be inaccessible in practice. The motivation for this design stems from the operational constraints inherent in real-world credit risk management, as data preparation, model execution, communication of results, decision-making, and institutional implementation all require time. Without such a buffer, the feature window would immediately precede the observation period, meaning that even if the model correctly predicts defaults within the observation period, by the time the above necessary steps are completed, the defaults may have already occurred. As a result, even accurate model predictions may become operationally unusable, leaving no opportunity for financial institutions to intervene. The blank interval therefore serves as a “no-input-data” buffer period, providing institutions with sufficient time to act upon the predictions, thereby enabling the development of a genuinely actionable early-warning model. This design is analogous to the notion of “lead time” in disaster warning systems [2]. Together, these two innovations support more accurate and operationally actionable several-month-ahead default prediction.

This study evaluates the proposed models under six experimental conditions. The first condition identifies the most effective resampling method to address class imbalance in the benchmark dataset utilized in this study. The second condition assesses model performance across different feature window lengths. The third condition conducts a sensitivity analysis on the impact of blank intervals, evaluating how intervals of 1–12 months affect model performance. The fourth condition investigates efficiency across varying sample sizes and aims to demonstrate that the proposed models can deliver comparable or superior performance even with smaller datasets, which represents a practical advantage for deployment. The fifth condition examines concept drift at both multi-year and quarterly levels and further explores model robustness on longer and shorter cycles of data when using the optimal resampling strategy, feature window, sample size, and blank interval identified in prior experiments. The sixth condition is an ablation study that illustrates the independent contribution of each model component. All conditions employ OOT test sets, where testing data originate from different years or quarters than training data and are entirely independent. This design demonstrates the models' ability to leverage extensive historical data, adapt to new data, and support near-real-time applications.

The primary contributions of this study are:

1. To propose innovative KAN-based GRU and LSTM models that effectively capture complex nonlinear temporal dynamics, enabling accurate and robust prediction of loan defaults several months in advance.
2. To propose the concept of a “blank interval” to simulate the real-world scenario of predicting loan defaults several months in advance under practical operational constraints, thereby enabling the development of an effective early-warning model for financial institutions.
3. To introduce the Transformer architecture into the context of loan default prediction and conduct comparative experiments to evaluate its ability to capture complex patterns and long-term dependencies, thereby addressing a gap in current research.
4. To provide comprehensive empirical evidence that clarifies the respective strengths of GRU-KAN and LSTM-KAN, thereby offering additional solution options for practical applications.

This study focuses on improving loan default prediction by overcoming the limitations of current models and enhancing their predictive accuracy. The resulting methodology enables financial institutions to identify risks earlier and take timely preventive actions, thereby strengthening their ability to mitigate potential losses.

2. Related works

2.1. Research gaps in early prediction of loan defaults

Previous studies on predicting loan defaults show differences and limitations in aspects such as test set construction and utilization of time series data. For example, Zandi et al. [3] used loan repayment data from loans issued in 2009 and 2010 in the Freddie Mac dataset, covering the period from January 2012 to June 2013, with a feature window every six months, totaling 13 windows for the training set; and used

the repayment data for July to December 2013 as the feature window for the test set to forecast defaults during the next 12 months. Being a test set derived from different segments of the same borrowers' repayment behavior time series as the training set, the patterns in the test data resemble those of the training set, potentially leading to overly optimistic prediction results. Furthermore, the prolonged time after loan issuance to attain the test set indicates that the study may not have fully addressed the practical applicability of the model. Breeden and Leonova [4] investigated the integration of machine learning with survival analysis to predict loan defaults. Using panel data from 1999 to 2019, the study used both OOS and OOT test sets over a 24-month observation period. However, the length of this period affected the model in accurately identifying if a default might occur sooner, such as within three months, thus limiting its usefulness in time-sensitive predictions. [5] examined loan default prediction but omitted details on handling time series data and test set design, viewing it as a static binary classification issue for unbalanced data. Similarly, Guo et al. [6] developed the HULHEN model to better profile loan applicants, improving the accuracy of default prediction, but also neglected time series data handling and experimental design. Lai [7] and Al-qerem et al. [8] also reduced the problem to a static binary classification, not adequately addressing temporal factors.

In summary, existing studies on loan default prediction have mainly used static data and ignored the ability of time series data to detect temporal patterns and trends. Even studies using time series data have mostly concentrated on OOS data, with OOT data being less utilized. Furthermore, few have attempted to integrate blank intervals to simulate "early prediction". These limitations have led to a poor understanding of the early prediction performance of existing models and a lack of exploration of the dynamic nature of time series data. To address these issues, our study aims to develop models using OOT data that excel in early prediction tasks, thereby improving the accuracy and practical applicability of loan default prediction.

2.2. Time series anomaly detection

Multiple empirical studies, such as Su et al. [9] and Touzani and Douzi [10], have shown that GRU and LSTM, although basic, remain among the leading models in the field of time series anomaly detection. Zamanzadeh Darban et al. [11] provided a comprehensive review of deep learning for time series anomaly detection across various domains, noting that GRU and LSTM are the most commonly used models in financial time series anomaly detection. Therefore, this study selects GRU and LSTM as a solid foundation for the proposed improvements. This section reviews research employing LSTM and GRU algorithms, with LSTM often preferred for capturing long-term temporal dependencies, while GRU, with its simpler structure, offers faster training times and is well-suited for efficiently modeling temporal dependencies.

2.2.1. GRU related applications

Liang and Cai [12] assessed the GRU model against LSTM, ARIMA, SVM, and ANN through two cross-validation techniques and three temporal dimensions using monthly loan default rates from the Lending Club P2P platform (2008-2015). The findings indicated better performance of LSTM in several metrics, while GRU effectively handles long-term dependencies with reduced computational demand, making it suitable for time series analysis. Hussein Sayed et al. [13] compared the prediction accuracy of five machine learning classifiers (Gaussian Naive Bayes, AdaBoost, Gradient Boosting, K-Nearest Neighbors, Decision Trees, Random Forest and Logistic Regression) and eight deep learning algorithms (multi-layer perceptron (MLP), convolutional neural networks (CNN), LSTM, Transformer, GRU, Autoencoder, ResNet, and DenseNet) to predict loan defaults using four metrics. Tang et al. [14] introduced GRN, an interpretable multivariate time series anomaly detection method utilizing Graph Neural Networks and GRU. GRU was selected for its proficiency in capturing long- and short-term dependencies in time series data with a lower computational cost than LSTM. Liu

et al. [15] developed a novel GRU-based anomaly detection approach for network logs, using the support vector domain description (SVDD). Experiments with the KDD Cup99 dataset indicated that it outperformed the classical GRU-MLP and LSTM methods. Similarly, Guo et al. [16] designed GGM-VAE, a GRU-based Gaussian mixture anomaly detection system, integrating GRU with Gaussian mixture priors to model multi-modal data distributions. The tests showed significant improvements in accuracy and F1 scores over traditional GRU models. These investigations highlighted the effectiveness of GRU-based designs in improving time series anomaly detection.

2.2.2. LSTM related applications

Benchaji et al. [17] utilized LSTM as a model to detect credit card fraud, focusing on historical purchase patterns to improve the detection accuracy of new fraudulent transactions. Gao et al. [18] created a rural microcredit risk assessment model based on LSTM, named the self-organizing LSTM algorithm, designed to capture the long-term dependencies between past actions and associated risk factors of borrowers. Menggang et al. [19] introduced the sliding window and the attention mechanism LSTM (SL-ALSTM), which combines LSTM with a sliding window and the attention mechanism. This model enhances LSTM's capacity to capture temporal dependencies by employing sliding windows for context extraction from loan data and utilizing the attention mechanism to prioritize significant data. This approach refines pattern recognition and time series forecasting, outperforming ARIMA, SVM, ANN, LSTM, and GRU models in the Lending Club public dataset. Benchaji et al. [20] developed a credit card fraud detection system using data sequence modeling, LSTM, and an attention mechanism, which assigns varying attention to information from the hidden layer of LSTM. This system considers the sequential structure of transactions, identifying crucial transactions in the input and improving the prediction of fraudulent activities. Experiments revealed that this model achieved the highest precision and recall compared to base models such as LSTM, GRU, and ANN. Ding et al. [21] introduced LGMAD, a real-time anomaly detection algorithm which was validated on the NAB dataset utilizing LSTM and a Gaussian mixture model (GMM). The anomalies in each univariate sensor time series are assessed with the LSTM, followed by multidimensional anomaly detection with the GMM.

In recent years, the emergence of Transformers has led to numerous studies exploring their use for time series data prediction [22]. However, opinions differ among scholars about the suitability of Transformers for these tasks. Su et al. [9] proposed WaveLST-Trans for anomaly detection in financial time series, specifically for cryptocurrency and stock index data containing both high-frequency and low-frequency components. The model combines wavelet transform, LSTM, and Transformer to capture multi-scale temporal patterns, improving detection accuracy and robustness compared with baseline models. Kim et al. [23] used Transformer models for innovating time series anomaly detection by simultaneously assessing global and local patterns. The model features an encoder with multiple Transformer layers and a decoder with one-dimensional convolutional layers; anomaly scores are the deviations between predicted and actual values at each timestamp. Benchmark tests show that the proposed approach exceeds the performance of LSTM and CNN, underscoring the effectiveness of the Transformer layers. Shi et al. [24] introduced a forecasting model that merges the Transformer self-attention mechanism with LSTM to efficiently capture long-term dependencies. To improve search efficiency, a hybrid of random search and Bayesian optimization refines parameters and regularization. The accuracy of the LSTM-Transformer model is validated by comparisons with LSTM, CNN, Transformer, and CNN-LSTM, showing that it achieves the best accuracy.

In contrast, Zeng et al. [22] questioned the effectiveness of popular Transformer-based methods in long-term time series forecasting (LTSF). The study used a straightforward linear model, LTSF-Linear, as the direct multi-step (DMS) benchmark to support this assertion. Importantly, the key contribution of the study is not the linear model

itself but the identification of a significant concern, as shown through comparative experiments, indicating that Transformers might not be universally effective as claimed. Although the study revealed important limitations of Transformers in certain long-horizon forecasting tasks, its findings do not imply that Transformer-based architectures are unsuitable for all categories of time series problems. In particular, research applying Transformers to loan default prediction in the context of time series anomaly detection remains limited. Given the capability of the self-attention mechanism to model complex temporal relationships and long-range dependencies, further exploration of Transformer-based models in this domain is warranted. This motivates the inclusion of Transformer architectures in the comparative experiments of this study to examine whether they can enhance the capture of intricate patterns relevant for several-month-ahead default prediction.

2.3. KAN related applications

KAN, pioneered by an MIT research team [25], is an innovative model that can potentially be used for detecting anomalies in time series. The method decomposes complex time series into univariate functions, replacing traditional linear weights with spline-parameterized alternatives, thus enabling dynamic learning of activation patterns and significantly enhancing interpretability. Somvanshi et al. [26] conducted a comprehensive survey on Kolmogorov-Arnold Networks, exploring the architectural advantages of KAN, including edge-based adaptive activation functions, and demonstrated its potential to improve time series prediction.

Multiple empirical studies have demonstrated the capabilities of KAN in time series anomaly detection. Das et al. [27] evaluated the effectiveness of KAN for anomaly detection in mechanical time series. The study compared KAN with traditional MLP-based neural networks and autoencoders, showing that KAN improves classification accuracy, training efficiency, generalization, continual learning, and resilience to catastrophic forgetting. Xu et al. [28] investigated how KAN can be used for time series forecasting, introducing T-KAN and MT-KAN. T-KAN is oriented to detect concept drift, using symbolic regression to explain non-linear links between predictions and previous time steps, ensuring

interpretability in dynamic contexts. MT-KAN improves forecasting by extracting and using complex connections between variables in multivariate time series. Experiments validated the efficacy of these methods, with T-KAN and MT-KAN significantly outperforming baseline models such as MLP, RNN, and LSTM in forecasting tasks. Vaca-Rubio et al. [29] explored the use of KAN in time series prediction using its adaptive activation functions to improve predictive modeling. The experiments showed that KAN surpassed traditional MLP in real-world satellite traffic forecasting with fewer learnable parameters and higher accuracy. Livieris [30] proposed a forecasting model called C-KAN for multistep predictions, where it combines convolutional layers with the KAN architecture. This model uses convolutional layers to capture the behavior and internal patterns of time series data, facilitating promising feature analysis and potentially more precise input-output management. Zhou et al. [31] introduced KAN-AD to address KAN's susceptibility to local anomalies caused by optimizing univariate functions with spline functions. KAN-AD emphasizes global patterns with Fourier series, reducing the impact of local peaks. Consequently, by transforming existing black-box learning methods into learning weights for univariate functions, it improves both effectiveness and efficiency.

KAN, considered an advanced MLP variant, has not yet been studied for the prediction of loan default. The originality and potential of such innovative algorithms in offering new insights justify exploring the application and effectiveness of KAN in this area, thus expanding its usage.

3. Methodology

3.1. Design of GRU-KAN and LSTM-KAN

This study explores the interaction between different feature extraction methods and the KAN model, proposing a hybrid architecture that integrates GRU-KAN and LSTM-KAN to enhance the ability to capture complex patterns in time series data. Specifically, both models share a similar architecture comprising a data preprocessing layer, a masking layer, a feature extraction layer, a KAN layer, a fully connected layer, and an output layer. The key difference between the two lies in the recurrent neural network layer used: GRU-KAN employs a GRU network,

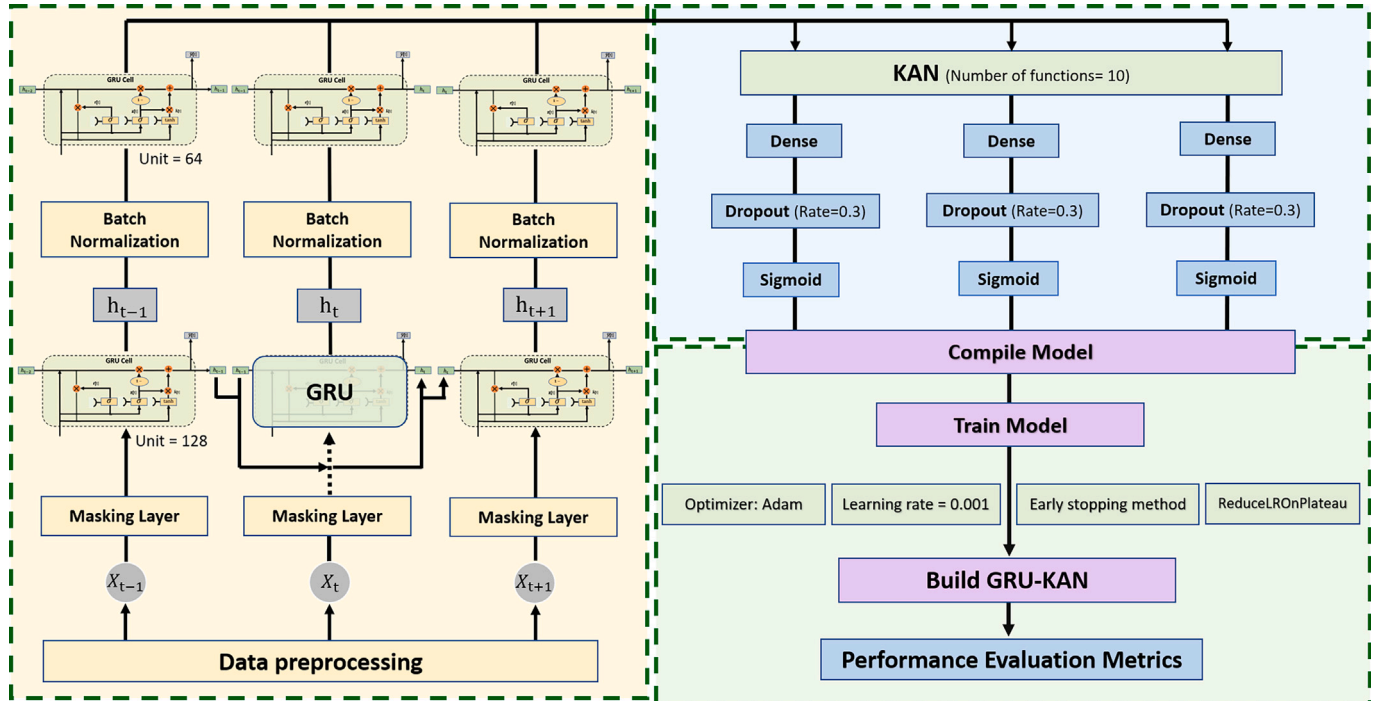


Fig. 1. Configuration of the proposed GRU-KAN.

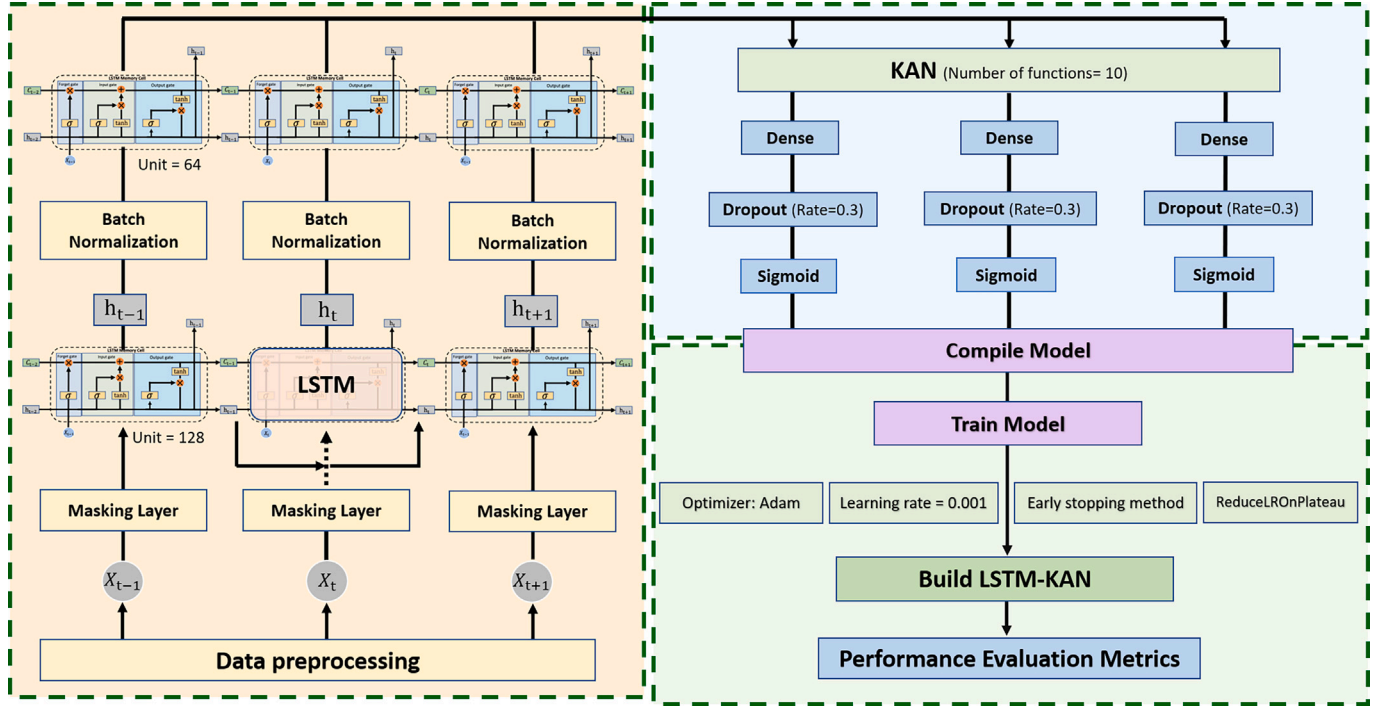


Fig. 2. Configuration of the proposed LSTM-KAN.

while LSTM-KAN uses an LSTM network. LSTM retains long-term dependencies effectively through its gating mechanism in the feature extraction layer, whereas GRU simplifies this process with fewer parameters, achieving a lightweight model while maintaining robust temporal modeling capabilities.

Figs. 1 and 2, along with the pseudocode in Algorithms 1 and 2, provide detailed descriptions of the GRU-KAN and LSTM-KAN architectures. The term “model” in the following definitions applies to both GRU-KAN and LSTM-KAN due to their shared structural characteristics.

Algorithm 1: Pseudocode of the proposed GRU-KAN.

Input: Data $X_i, 1 \leq i \leq N$

- 1 Initialize model $M(\text{GRU}, \text{KAN})$ with parameters θ
- 2 **for each** input sequence X_i **for** $i = 1, \dots, N$ **do**
 - $X' \leftarrow \text{Preprocess}(X_i, \theta_{\text{masking}})$ $\text{Masked}_{\text{input}} \leftarrow \text{Masking}(\text{mask_value} = 0.0)(X')$
 - $h_{\text{gru1}} \leftarrow \text{GRU}(\text{units} = 128, \text{return_sequences} = \text{True}, \theta_{\text{gru}})(\text{Masked}_{\text{input}})$
 - $h_{\text{gru1_norm}} \leftarrow \text{BatchNormalization}(\theta_{\text{bn}})(h_{\text{gru1}})$ $h_{\text{gru2}} \leftarrow \text{GRU}(\text{units} = 64, \text{return_sequences} = \text{False}, \theta_{\text{gru}})(h_{\text{gru1_norm}})$ $\text{kan}_{\text{out}} \leftarrow \text{KAN}(\text{output_dim} = 1, \text{num_functions} = 10, \theta_{\text{kan}})(h_{\text{gru2}})$ $\text{dense}_{\text{out}} \leftarrow \text{Dense}(\text{units} = 64, \text{activation} = \text{'relu'}, \theta_{\text{dense}})(\text{kan}_{\text{out}})$ $\text{dropout}_{\text{out}} \leftarrow \text{Dropout}(\text{rate} = 0.3, \theta_{\text{dropout}})(\text{dense}_{\text{out}})$ $\text{instance}_{\text{output}} \leftarrow \text{Dense}(\text{units} = 1, \text{activation} = \text{'sigmoid'}, \theta_{\text{output}})(\text{dropout}_{\text{out}})$ **Return** $\text{instance}_{\text{output}}$

Output: $M(\text{GRU}, \text{KAN}, \theta_{\text{output}}), \text{instance}_{\text{output}}$

Algorithm 2: Pseudocode of the proposed LSTM-KAN.

Input: Data $X_i, 1 \leq i \leq N$

- 1 Initialize model $M(\text{LSTM}, \text{KAN})$ with parameters θ
- 2 **for each** input sequence X_i **for** $i = 1, \dots, N$ **do**
 - $X' \leftarrow \text{Preprocess}(X_i, \theta_{\text{masking}})$ $\text{Masked}_{\text{input}} \leftarrow \text{Masking}(\text{mask_value} = 0.0)(X')$
 - $h_{\text{lstm1}} \leftarrow \text{LSTM}(\text{units} = 128, \text{return_sequences} = \text{True}, \theta_{\text{lstm}})(\text{Masked}_{\text{input}})$
 - $h_{\text{lstm1_norm}} \leftarrow \text{BatchNormalization}(\theta_{\text{bn}})(h_{\text{lstm1}})$ $h_{\text{lstm2}} \leftarrow \text{LSTM}(\text{units} = 64, \text{return_sequences} = \text{False}, \theta_{\text{lstm}})(h_{\text{lstm1_norm}})$ $\text{kan}_{\text{out}} \leftarrow \text{KAN}(\text{output_dim} = 1, \text{num_functions} = 10, \theta_{\text{kan}})(h_{\text{lstm2}})$ $\text{dense}_{\text{out}} \leftarrow \text{Dense}(\text{units} = 64, \text{activation} = \text{'relu'}, \theta_{\text{dense}})(\text{kan}_{\text{out}})$ $\text{dropout}_{\text{out}} \leftarrow \text{Dropout}(\text{rate} = 0.3, \theta_{\text{dropout}})(\text{dense}_{\text{out}})$ $\text{instance}_{\text{output}} \leftarrow \text{Dense}(\text{units} = 1, \text{activation} = \text{'sigmoid'}, \theta_{\text{output}})(\text{dropout}_{\text{out}})$ **Return** $\text{instance}_{\text{output}}$

Output: $M(\text{LSTM}, \text{KAN}, \theta_{\text{output}}), \text{instance}_{\text{output}}$

The data pre-processing layer is responsible for preparing loan-related input data before feeding it into the model. A detailed description of this process can be found in Section 3.2.2. The masking layer identifies padding values in the input data. Since loan-related time series data often have varying sequence lengths, shorter sequences are typically padded to match the longest sequence for batch processing. The masking layer distinguishes between actual data and padding values, ensuring the model focuses only on meaningful inputs while ignoring artificial padding. This step prevents training interference from filler values, allowing the model to capture essential patterns more effectively and improve prediction accuracy.

After masking, the data is passed to the recurrent layer for feature extraction. Two specific recurrent architectures, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are applied and compared. Both LSTM and GRU are variants of Recurrent Neural Networks (RNNs) and are designed to effectively handle time series data by capturing dependencies across different time steps. For example, in loan data, a borrower's repayment behavior at different time points is inherently correlated. By leveraging LSTM or GRU layers, the model can learn these temporal dependencies and extract dynamic features related to loan default risks.

In this study, both models employ a two-layer feature extraction structure, where the sequential data is first processed through two stacked LSTM or GRU layers. To improve training stability and generalization, a Batch Normalization layer is introduced between the two feature extraction layers.

Feature Extraction Stage:

LSTM introduces a gating mechanism and memory cells to mitigate the vanishing gradient problem, thereby slowing down memory decay during back-propagation. LSTM uses a set of gates to regulate the flow of information. The LSTM cell first processes information through the forget gate [32]:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (1)$$

where σ is the sigmoid activation function, and W_f, b_f are trainable parameters.

After the forget gate, the LSTM cell evaluates new information through the input gate and the candidate cell state. The input gate is defined as

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (2)$$

while the candidate cell state is computed as

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (3)$$

The input gate regulates how much new information is added to the memory cell, while the candidate cell state represents the new memory content that could be incorporated. The cell state is then updated as a combination of the old state, filtered by the forget gate, and the new candidate state. The update process follows the equation

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (4)$$

where $\odot$ represents element-wise multiplication. This step ensures that relevant past information is preserved while new meaningful data is integrated into the memory. Finally, the output gate determines the hidden state, which serves as both the short-term memory and the output of the LSTM cell. This is calculated using the formula

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (6)$$

The output gate decides how much of the cell state should be exposed, while the hidden state serves as the output for the current step and as input for the next step.

When the feature extraction layer consists of LSTM layers, it includes two key components: the first LSTM layer with 128 units and the second LSTM layer with 64 units. Assume that the masked input time series is $Masked_{input} = \{x_1, x_2, \dots, x_T\}$.

The first LSTM layer is structured to output a sequence of hidden states $h_{lstm1} = \{h_1, h_2, \dots, h_T\}$ corresponding to each time step, rather than a single summary representation. This design choice allows the model to retain the temporal structure of the input data, ensuring that the extracted features capture both short-term and long-term dependencies.

Following the first LSTM layer, a Batch Normalization (BatchNorm) layer is applied to stabilize training and improve generalization. This normalization process is governed by the following formula [33]:

$$h_{norm} = \gamma \frac{h_{lstm1} - \mu}{\sigma + \epsilon} + \beta \quad (7)$$

where:

- μ and σ are the mean and standard deviation computed over the batch,
- γ and β are learnable parameters,
- ϵ is a small constant added for numerical stability,
- The arithmetic operations are element-wise.

By adjusting the mean to zero and the variance to one, Batch Normalization effectively mitigates the issues of vanishing or exploding gradients, leading to a more stable optimization process. In addition, it accelerates convergence, enabling faster training and improved overall performance of the model. After passing through the first LSTM layer and the Batch Normalization layer, the data are transformed into a stable and feature-rich representation. To further refine these extracted features, a second LSTM layer is introduced, which selectively emphasizes the most critical information through its gating mechanisms. This layer focuses on capturing the most salient temporal dependencies from the normalized sequence, ensuring that only the most relevant patterns are retained for subsequent processing. Unlike the first LSTM layer,

which preserves the full temporal structure, the second LSTM layer is designed to summarize the learned information in a single hidden state by returning only the hidden representation of the final time step. This compressed output forms the final feature vector $h_{lstm2} \in R^{64}$, where 64 represents the number of hidden units in the second LSTM layer.

GRU simplifies the structure of LSTM while effectively addressing the vanishing gradient problem. Unlike standard RNNs, which struggle with long-term dependencies due to gradient decay, GRU introduces two gates: the reset gate and the update gate. These gates regulate the flow of information, allowing the model to retain essential historical data while discarding irrelevant past information. Compared to LSTM, GRU has fewer parameters and is computationally more efficient while maintaining comparable performance in many sequence modeling tasks. The update gate plays a crucial role in determining how much of the previous hidden state h_{t-1} should be carried forward to the current hidden state h_t . It helps the model decide whether to retain past information or update it with new input from the current time step. The update gate is computed as follows:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad [12] \quad (8)$$

where:

- x_t represents the input at time step t .
- h_{t-1} is the hidden state of the previous time step.
- W_z and U_z are weight matrices.
- b_z is the bias term.
- $\sigma(\cdot)$ is the sigmoid activation function.

The reset gate of GRU controls how much of the past hidden state should be forgotten before computing the new candidate hidden state. The reset gate is defined as:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad [12] \quad (9)$$

where r_t determines how much the hidden state of the past contributes to the current computation. Once the reset gate has filtered the information, a candidate hidden state $\tilde{h}_t$ is computed using:

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad [12] \quad (10)$$

The final hidden state h_t is a weighted combination of the old hidden state h_{t-1} and the candidate hidden state $\tilde{h}_t$, modulated by the update gate:

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t \quad [12] \quad (11)$$

where z_t indicates whether the model updates with new information or retains past information.

In the feature extraction layer, the GRU structure consists of two layers, similar to the LSTM design: the first GRU layer with 128 processing units and the second with 64 units. The hidden state from the first GRU layer is passed through batch normalization, while the second GRU layer delivers the final feature vector to the KAN layer.

Nonlinear Modeling Stage:

The key idea behind Kolmogorov-Arnold Networks (KAN) is that any multi-variable function $f(x_1, \dots, x_n)$ can be transformed into a combination of multiple single-variable functions. By leveraging non-linear transformation functions, KAN is capable of capturing intricate non-linear interactions between different variables, which traditional linear models may struggle to represent effectively.

In this model, LSTM or GRU serves as the feature extraction layer, producing a hidden state vector that encodes temporal dependencies from sequential data. However, the relationships between these hidden state features are often highly complex and non-linear. For instance, in the context of loan default prediction, factors such as a borrower's credit

rating, income level, and debt status do not exhibit a simple linear correlation with the likelihood of default. Instead, these factors interact in a more intricate manner, requiring a model that can effectively decompose and understand their underlying dependencies.

KAN addresses this challenge by decomposing the complex relationships embedded in the hidden state vector. KAN layer follows sequence processing, specifically targeting the output of the second GRU and LSTM layers. Unlike MLPs, which employ fixed node activation functions, KAN utilizes edge-based learnable activation functions [25]. Specifically, each weight in an MLP is replaced by a univariate function parameterized as a spline function, thus eliminating the need for linear weights in the network. In particular, a conventional “input-hidden-output” structure of an MLP can be substituted by the composition of two KAN layers, satisfying the following equation:

$$KAN_{output} = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{p,q}(x_p) \right) \quad [25] \quad (12)$$

where

- x_p represents the p -th element of the hidden state vector.
- $\phi_{p,q} : [0, 1] \rightarrow \mathbb{R}$ is a learned non-linear function that transforms each individual feature separately.
- $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$ is another non-linear activation function, applied to the aggregated sum to introduce additional flexibility in modeling complex interactions.

When extending a single KAN layer into L-stacked layers, the model can be represented in a more compact form as:

$$KAN_{output} = (\Phi_L \circ \Phi_{L-1} \circ \dots \circ \Phi_1)x \quad [25] \quad (13)$$

In this hierarchical KAN network, each individual KAN layer functions as a small sub-network, progressively transforming the input through multiple levels of non-linear feature extraction. By stacking multiple KAN layers, the model gains enhanced expressive power, enabling it to capture increasingly complex relationships and deeper hierarchical representations within the data. In this study, a single KAN layer (i.e., $L = 1$) is utilized, as the structural simplicity of the dataset does not warrant the use of deeper KAN stacking. After the KAN layer, the fully connected layer performs a non-linear transformation on the univariate functions output by the KAN layer. By applying linear transformations and activation functions, the model further processes and refines the extracted features, enabling the model to learn higher-level feature representations.

To enhance generalization and prevent overfitting, a regularization layer, specifically a dropout layer, is introduced to randomly deactivate a portion of neurons during training. This prevents the model from over-relying on specific neurons, reducing the risk of overfitting, where the model performs well on the training set but struggles with new or unseen data. The output layer will transform the output of the fully connected layer into a probability value ranging from 0 to 1, representing the predicted probability of default.

3.2. Dataset and data preprocessing

3.2.1. Dataset

The experiment utilizes the Single-Family Loan-Level dataset from Freddie Mac [34]. This dataset includes comprehensive details on loan dates, overdue status, and vital information for fraud detection. Each yearly or quarterly dataset is structured as panel data comprising the information of borrowers who applied for loans during that year or quarter. The data for each borrower consist of a multidimensional time series, including monthly repayment records and the corresponding values of all features for that month. The target variable, known as default, is binary, based on the current loan delinquency status (CLDS), indicating days overdue since the last installment was due. A CLDS value of 3 or more results in a default value of 1; otherwise it is 0. This

Table 1

Overview of feature descriptions in the Single-Family Loan-Level dataset by Freddie Mac [34].

Feature	Description
Target variable: Default	Equals 1 when CLDS is 3 or more, and 0 otherwise.
Input variables: Assistance status code	Type of support arrangement for short-term loan payment relief.
Current actual UPB	Indicates the stated final unpaid principal balance (UPB) of the mortgage.
Current deferred UPB	The present non-interest accruing UPB of the adjusted loan.
Current interest rate	Displays the present interest rate on the mortgage note, including any modifications.
Estimated loan to value	Present LTV ratio as determined by Freddie Mac's AVM value.
Interest bearing UPB	The present UPB of the adjusted loan that accumulates interest.

Table 2

Statistics of datasets extracted from the Freddie Mac Single-Family Loan-Level dataset [34].

Year	Training Set	Test Set	Sample Size	No. of Loans	Average Loan Length (Months)	Default Rate (%)	No. of Default
2018	✓		44,481,890	1,231,614	36.12	5.27	64,960
2019	✓	✓	58,076,701	1,716,106	33.84	5.14	88,160
2020	✓	✓	164,170,587	3,810,240	43.09	1.69	64,209
2021	✓	✓	160,196,538	3,960,435	40.45	1.64	65,025
2022		✓	46,913,011	1,508,832	31.09	2.55	38,475

default definition follows the Basel II industry standard [35]. To preserve the practical significance of the task, samples exhibiting nonzero CLDS values within the input feature window are excluded prior to the training-testing split, thereby ensuring that the model focuses on genuine early-stage prediction rather than merely identifying already evident delinquency signals. Six features per transaction are selected as input variables, detailed in Table 1.

Four consecutive training-testing configurations are formed: 2018 data are used for training, while 2019 data are used for testing; 2019 data are used for training and 2020 data for testing; 2020 data are used for training and 2021 data for testing; 2021 data are used for training and 2022 data for testing. Table 2 presents the summary statistics of the original dataset, where sample size denotes the total number of monthly records across all borrowers (reflecting the raw panel structure), the number of loans represents the number of distinct loan sequences (each corresponding to one borrower's loan tracked over time), the average loan length measures the temporal length of these sequences in months, and the number of defaults and default rate respectively indicate the total number and the proportion of loans that defaulted.

To ensure computational feasibility given the large scale of the raw data, each yearly dataset is formed by stratified random sampling of one-tenth of the records from each quarter of the original dataset, preserving the distribution of key variables. In addition, for further evaluation, this study also constructs four datasets with different imbalance ratios from the original test data, which is characterized by a highly imbalanced distribution. These datasets serve as test sets, resulting in a total of 20 test sets. Table 3 reports the detailed statistics of the 20 datasets. The default ratios of 1:1, 1:2, 1:5, and 1:10 are obtained by retaining all default samples and randomly undersampling the non-default samples at different proportions, so that the only difference from the original dataset of the same year lies in the composition of the non-default samples. The number of defaults refers to the total defaults without restricting the input window length. In the experiments presented in Section 4, the specified input window length, blank interval and observation period are

Table 3
Constructed test sets with different imbalance ratios.

Test set	Year	Default:Non-default	No. of Defaults	No. of Non-default
D1	2019	1:1	8786	8786
D2	2019	1:2	8786	17,572
D3	2019	1:5	8786	43,930
D4	2019	1:10	8786	87,860
D5	2019	1:18.53 (original)	8786	162,823
D6	2020	1:1	6376	6376
D7	2020	1:2	6376	12,752
D8	2020	1:5	6376	31,880
D9	2020	1:10	6376	63,760
D10	2020	1:58.76 (original)	6376	374,646
D11	2021	1:1	6477	6477
D12	2021	1:2	6477	12,954
D13	2021	1:5	6477	32,385
D14	2021	1:10	6477	64,770
D15	2021	1:60.15 (original)	6477	389,564
D16	2022	1:1	3844	3844
D17	2022	1:2	3844	7688
D18	2022	1:5	3844	19,220
D19	2022	1:10	3844	38,440
D20	2022	1:38.25 (original)	3844	147,038

explicitly stated, and the corresponding number of defaults represents the count under these constraints.

3.2.2. Data preprocessing

The dataset is structured by loan sequence number, with data sorted by the remaining months to legal maturity to satisfy the window size and prediction period requirements. The window length is defined on the basis of the test condition, and sequences shorter than this length are removed. The borrower assistance status code is transformed using One-Hot encoding to convert the categorical variable into a binary vector suitable for model processing. A new feature, known as interest-bearing UPB-delta, is derived by calculating the first-order difference of the interest-bearing UPB, highlighting the changes between consecutive points. This feature replaces the original interest-bearing UPB to reduce long-term trends and seasonal effects, using first-order differencing to emphasize short-term variations, which is a widely used technique in feature engineering.

Six resampling methods were evaluated to address class imbalance, which significantly impacts the performance and generalization of machine learning models, especially in real-world applications such as loan default prediction. The methods include random undersampling (RUS), random oversampling (ROS), Synthetic Minority Over-sampling Technique (SMOTE), tSMOTE [36], TimeGAN [37], and a baseline without resampling. Resampling methods were applied exclusively to the training sets. The RUS procedure retained all the default (minority class) instances, ensuring sample differences across 20 trial repetitions through the random selection of equal numbers of non-default (majority class) instances. In the ROS, SMOTE, tSMOTE, and TimeGAN methods, the majority class was first randomly undersampled to achieve a default-to-non-default ratio of 1:2, after which the minority class was oversampled to match the resulting number of majority class instances. This two-step approach was adopted because the original imbalance was excessively high, such that direct oversampling would have produced an impractically large dataset, whereas combining undersampling with oversampling generally yields improved classification performance.

3.3. Evaluation metrics

The performance of the model is assessed through standard metrics such as accuracy, precision, recall, F1, and AUC, offering a thorough evaluation of classification effectiveness. These metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (14)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (15)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (16)$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Accuracy quantifies the ratio of correctly predicted instances among all instances, providing a basic evaluation of the performance of the model. However, in imbalanced datasets, accuracy can be misleading as it often highlights the dominance of the majority class over true predictive capability. Precision denotes the fraction of true positive predictions among all predicted positives, whereas recall indicates the fraction of true positives correctly identified out of all actual positives. Ideally, a robust model achieves high precision and recall, but there is often a trade-off between them in practice. High precision suggests a cautious model that predicts positive only with high confidence, potentially missing true positives, and thus reducing recall. However, high recall might result from predicting more positives to capture as many true positives as possible, leading to more false positives and reducing precision. This trade-off highlights the necessity to balance these metrics, where focusing on one metric alone might not produce an ideal model.

F1 score (harmonic mean of precision and recall) provides a more comprehensive assessment. The classification threshold of F1 is set at 0.5, indicating a balanced class distribution. AUC (Area Under the Curve) assesses the class differentiation ability of a model by summarizing the trade-off between true and false positive rates at various thresholds. An AUC near 1 shows better discrimination. These metrics ensure that both positive and negative classes are addressed, especially in contexts where class trade-offs are critical. Although all five metrics are reported to provide a complete description of model performance, the subsequent multi-metric aggregated evaluation relies on a selected subset of these metrics.

Given the potential for divergent results between different evaluation metrics, this study applies a comprehensive evaluation method that integrates multiple metrics to obtain a more balanced assessment. This method follows the ranking-based aggregation principle and the *AvgR* (average ranking) concept introduced by Lessmann et al. [38], where model performance is represented by the *AvgR*, obtained by averaging the respective ranks assigned under each evaluation metric. To align the ranking-based evaluation method with the objectives of this study, *AvgR* is computed using the rankings derived from AUC, F1, and recall, rather than the specific metric combinations used in the cited study. These three metrics are selected in view of their importance in loan default prediction, particularly considering that within the broader context of financial anomaly detection, high recall, which reflects the ability to correctly identify true positives (TP), is typically more important than achieving high precision, since the cost of false negatives (FN, undetected risk) generally exceeds that of false positives. Overall, the *AvgR* used in this study represents an application-driven instantiation of an established evaluation methodology rather than a new mathematical metric. Subsequent subsections present task-specific *AvgR* variants corresponding to the experimental settings, together with their detailed calculation procedures (see Section 4).

4. Experiment design and results analysis

Six experiments are designed to evaluate the performance of the proposed models in six different aspects, including the impact of different resampling methods (see Section 4.1), the impact of different feature window lengths (see Section 4.2), the early prediction capability for default events (see Section 4.3), performance with limited training samples (see Section 4.4), model generalization performance analysis (see Section 4.5) and ablation study (see Section 4.6). The

evaluation involved comparing the proposed model against the baselines of GRU, LSTM, LSTM-Attention, and LSTM-Transformer. To ensure fairness, all models were tuned using a randomized search [39] with cross-validation over key hyperparameters. Validation sets drawn from independent data covering 2019 to 2022 (and different from the test sets) were used to select the configurations that achieved the best performance in most validation folds. Table A1 in the appendix presents the selected configurations and their detailed parameter ranges.

4.1. Resampling methods performance analysis

In this subsection, feature window lengths of 12 and 18 were adopted, as they are commonly employed in studies on credit default prediction and time series modeling. No blank interval is set in this subsection. The average predictive performance is expressed in terms of rankings, where smaller values indicate better performance, with 1 denoting the best result, represented by $AvgR_R$, which is defined as follows.

Let $\mathcal{D}$ denote the set of datasets ($|\mathcal{D}| = 20$), $\mathcal{R}$ ($|\mathcal{R}| = 6$) the set of resampling methods applied to the training set, and $\mathcal{M}$ the models including two proposed and four baselines ($|\mathcal{M}| = 6$). Let $recall_{d,r,m}$, $F1_{d,r,m}$ and $AUC_{d,r,m}$ represent the mean recall, F1 and AUC of 20 independent trials of applying the resampling method $r \in \mathcal{R}$, and model $m \in \mathcal{M}$ to dataset $d \in \mathcal{D}$. $recall_RANK_{R,M}(d, r, m)$, $F1_RANK_{R,M}(d, r, m)$ and $AUC_RANK_{R,M}(d, r, m)$ indicate their respective global rankings among all the corresponding metrics computed across all resampling methods and models in each dataset, respectively. Then, the total average ranking of d when r and m are applied, denoted as $AvgR_{R,M}(d, r, m)$, is given as:

$$AvgR_{R,M}(d, r, m) = \frac{AUC_RANK_{R,M}(d, r, m) + F1_RANK_{R,M}(d, r, m) + recall_RANK_{R,M}(d, r, m)}{3} \quad [45] \quad (18)$$

where $d \in \mathcal{D}$, $r \in \mathcal{R}$, and $m \in \mathcal{M}$.

Subsequently, these values are grouped based on different resampling methods given by the set $\mathcal{R} = \{ \text{RUS, ROS, SMOTE, tSMOTE, TimeGAN, and no resampling} \}$ with their within-group averages denoted as $AvgR_R$:

$$AvgR_R(d, r) = \frac{1}{|M(r)|} \sum_{m \in M(r)} AvgR_{R,M}(d, r, m) \quad (19)$$

where $r \in \mathcal{R}$ and $M(r) \subset \mathcal{M}$ denotes the index set of models using resampling method r .

Tables 4 and 5 report the impact of different resampling strategies on the average predictive performance of six models across 20 datasets, for feature window lengths of 12 and 18, respectively. It can be observed that, under the experimental setup of this study, RUS consistently achieves the best $AvgR_R$ across the 20 datasets, regardless of whether the feature window length is 12 or 18, followed closely by tSMOTE and TimeGAN. Consequently, subsequent experiments adopt RUS as the sole resampling strategy to ensure both efficiency and robustness in model evaluation.

Moreover, feature window length of 18 generally yields higher recall, F1, and AUC than a length of 12 in most datasets. Specifically, due to the large amount of data, Fig. 3 uses dataset D1 as an example to present heatmaps of the 20-trial average F1 and AUC for the six models, based on a feature window length of 18 and a 3-month observation period as the target variable y .

As illustrated by the heatmap, the performance differences induced by different resampling methods within the same model are generally minor, suggesting that model outcomes are not highly sensitive to the choice of resampling strategy. Regardless of the resampling method applied, our proposed GRU-KAN and LSTM-KAN models consistently outperform the baseline models across the key evaluation metrics.

Table 4

$AvgR_R(d, r)$ of the 20 datasets under different oversampling methods when feature window length is 12 (lower is better).

Dataset	RUS	tSMOTE	TimeGAN	SMOTE	ROS	No Resampling
D1	17.2	17.5	18.5	19.3	18.4	20.1
D2	16.8	18.6	17.9	18.7	19.2	19.8
D3	18.1	18.1	17.5	18.8	17.8	20.7
D4	17.8	17.2	17.6	19.5	19.1	19.8
D5	17.3	18.6	17.9	18.8	18.2	20.2
D6	16.8	17.0	18.9	19.8	18.4	20.1
D7	18.2	18.4	18.0	19.2	17.7	19.5
D8	17.7	17.5	18.2	19.1	18.0	20.5
D9	18.3	18.1	17.4	18.6	19.1	19.5
D10	16.3	19.1	18.3	19.2	18.1	20.0
D11	17.1	19.1	17.8	18.8	18.3	19.9
D12	17.6	16.6	18.5	19.4	18.6	20.3
D13	16.7	18.5	18.7	18.9	17.5	20.7
D14	17.5	18.1	18.1	18.8	18.3	20.2
D15	17.9	17.4	17.9	18.9	18.4	20.5
D16	17.8	18.5	18.6	19.2	18.1	18.8
D17	17.1	18.8	18.2	19.1	17.6	20.2
D18	17.7	17.9	17.4	18.7	18.1	21.2
D19	17.3	16.9	18.3	18.8	18.9	20.8
D20	17.5	17.9	18.8	19.6	17.6	19.6
Average	17.6	18.0	18.1	19.1	18.2	20.1

Table 5

$AvgR_R(d, r)$ of the 20 datasets under different resampling methods when feature window length is 18 (lower is better).

Dataset	RUS	tSMOTE	TimeGAN	SMOTE	ROS	No Resampling
D1	16.3	17.4	18.0	18.6	18.9	21.8
D2	16.9	15.9	16.4	18.9	19.5	23.4
D3	17.4	17.9	18.3	18.2	17.8	21.4
D4	17.5	16.2	16.9	19.6	17.1	23.7
D5	15.1	17.3	16.0	18.3	18.4	25.9
D6	15.9	17.2	17.5	19.8	18.7	21.9
D7	16.4	17.3	17.8	17.6	20.6	21.3
D8	15.7	15.2	16.2	19.9	21.8	22.2
D9	15.4	15.8	16.9	19.4	20.9	22.6
D10	14.9	16.1	15.5	18.2	20.9	25.4
D11	15.3	15.8	16.4	19.1	21.6	22.8
D12	15.1	16.2	15.6	19.8	21.8	22.5
D13	16.2	17.2	17.6	17.9	20.2	21.9
D14	15.2	17.3	16.4	18.6	21.1	22.4
D15	14.4	17.3	17.5	19.1	18.5	24.2
D16	15.6	16.1	14.9	19.5	21.5	23.4
D17	16.1	15.6	16.1	17.4	22.2	23.6
D18	16.5	17.5	16.7	18.2	20.7	21.4
D19	15.8	16.4	16.2	19.5	20.2	22.9
D20	14.6	15.7	16.3	17.2	21.8	25.4
Average	15.8	16.6	16.7	18.7	20.2	23.0

4.2. Feature window lengths performance analysis

To assess how different feature window lengths affect the proposed model, the lengths of x are set to 12, 15, 18, 21, 24, and 27 months. These windows serve as inputs to predict defaults within the next 3 months, with this 3-month observation period acting as the target variable y . No blank interval is set in this subsection. For each feature window length, the models are trained using historical data from the months specified prior to the observation period. This analysis reveals the trade-off between feature window length and model performance, highlighting the optimal feature selection configuration in predictive tasks, helping us to understand the extent of the model's dependence on the feature window length and its ability to adapt on data with different time spans.

Let $\mathcal{L}$ denote the set of feature window lengths ($|\mathcal{L}| = 6$). The definitions of $\mathcal{D}$ and $\mathcal{M}$ are the same as in Section 4.1. Let $recall_{d,l,m}$, $F1_{d,l,m}$ and $AUC_{d,l,m}$ represent the mean recall, F1, and AUC obtained from 20 independent trials when applying feature window length $l \in \mathcal{L}$ and model $m \in \mathcal{M}$ to dataset $d \in \mathcal{D}$.

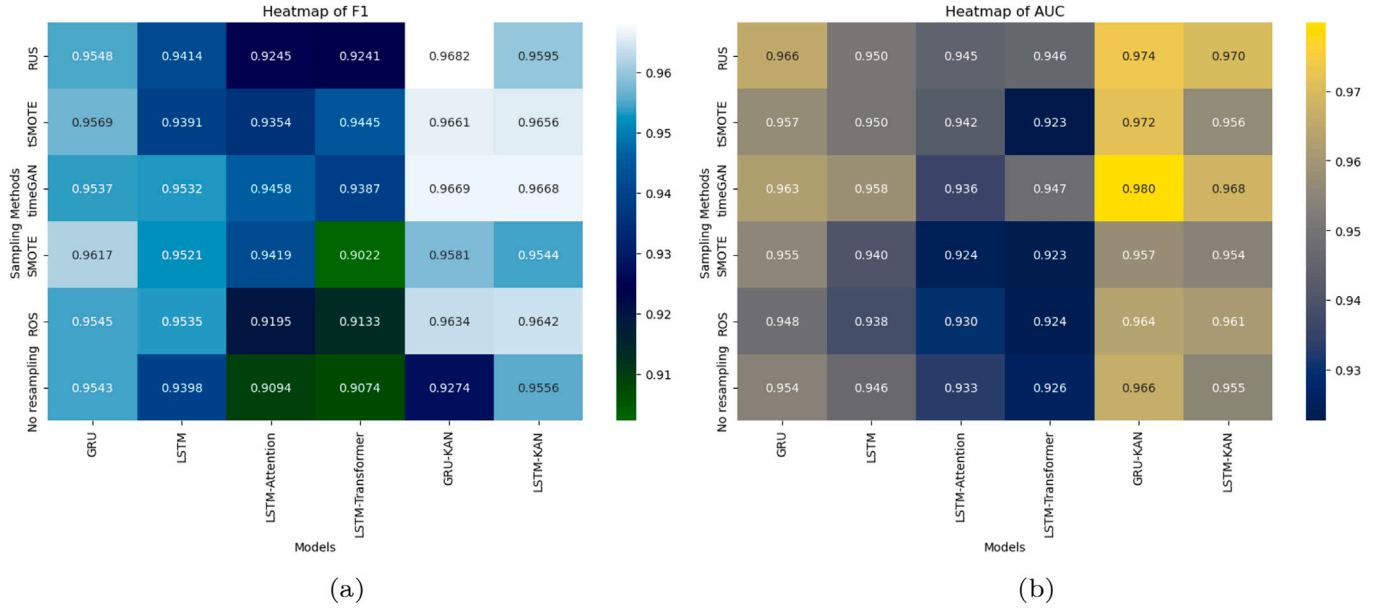


Fig. 3. Heatmap depicting (a) F1 and (b) AUC for each model using different resampling methods on D1.

Correspondingly, $recall_RANK_{L,M}(d, l, m)$, $F1_RANK_{L,M}(d, l, m)$, and $AUC_RANK_{L,M}(d, l, m)$ indicate their respective global rankings across all feature window lengths and models within each dataset.

The total average ranking for dataset d when using feature window length l and model m , denoted as $AvgR_{L,M}(d, l, m)$, is defined as:

$$AvgR_{L,M}(d, l, m) = \frac{AUC_RANK_{L,M}(d, l, m) + F1_RANK_{L,M}(d, l, m) + recall_RANK_{L,M}(d, l, m)}{3} \quad [45] \quad (20)$$

where $d \in D$, $l \in \mathcal{L}$, and $m \in \mathcal{M}$.

Subsequently, these values are grouped according to the feature window lengths in $\mathcal{L} = \{12, 15, 18, 21, 24, 27\}$, and their within-group averages are denoted as $AvgR_L$:

$$AvgR_L(d, l) = \frac{1}{|M(l)|} \sum_{m \in M(l)} AvgR_{L,M}(d, l, m) \quad (21)$$

where $l \in \mathcal{L}$, and $M(l) \subset \mathcal{M}$ denotes the set of models corresponding to feature window length l .

Table 6 illustrates the variation in predictive performance across the datasets when different feature window lengths are applied, considering multiple evaluation metrics and models. Lower values indicate better performance. It can be observed that models with a feature window length of 12–18 months perform better than those with other lengths, which is consistent with the findings of Kennedy et al. [40]. This trend can be understood in relation to the loan cycle. When the feature window is short, for example, less than 12 months, the patterns of borrowers' repayment behavior may not have fully appeared, and the signals of default or prepayment risk may not yet be evident. This results in insufficient feature discriminability and limits the predictive capacity of the model. In contrast, when the window is slightly longer, like 18–24 months, early repayment behaviors may deviate from more recent patterns, thus introducing noise and reducing predictive accuracy. As the window length extends further, performance shows a slight recovery, potentially due to the capture of broader macroeconomic and market trends, which can partially offset the negative effects of noise. In general, the optimal window length is more closely related to the data distribution. For example, datasets from 2019 (D1–D5) may be better

Table 6

$AvgR_L(d, l)$ of the 20 datasets under different feature window lengths (lower is better).

Dataset	12	15	18	21	24	27
D1	16.4	11.2	9.1	24.6	26.5	23.2
D2	17.1	9.6	10.1	25.2	26.3	22.7
D3	13.4	11.5	9.7	26.5	25.4	24.5
D4	15.8	9.3	11.2	24.9	26.1	23.7
D5	16.8	9.7	11.4	24.7	26.6	21.8
D6	10.7	8.7	16.7	24.7	26.2	23.9
D7	11.9	10.8	12.5	25.3	26.1	24.4
D8	12.8	9.5	15.1	24.6	26.5	22.5
D9	12.1	9.9	16.4	24.9	26.5	21.2
D10	11.9	10.7	12.2	25.2	26.2	24.8
D11	16.5	11.6	10.3	24.8	26.7	21.1
D12	16.3	9.6	11.4	24.9	26.3	22.5
D13	13.1	12.5	10.9	24.7	25.7	24.1
D14	17.6	11.7	9.5	24.7	26.2	21.3
D15	16.9	9.4	11.1	25.1	26.1	22.4
D16	12.4	14.5	11.3	21.5	27.2	24.1
D17	11.8	15.3	11.4	21.7	26.5	24.3
D18	11.1	13.9	12.9	22.7	25.3	25.1
D19	12.3	13.7	8.3	24.7	26.4	25.6
D20	11.1	14.3	9.8	21.9	29.1	24.8
Average	14.2	11.4	11.6	24.4	26.4	23.4

suit to feature windows of 15–18 months, while datasets from later years (D6–D10) might perform better with windows of 12–15 months. Generally, a feature window of 12–18 months provides an appropriate balance between stability and information sufficiency. This window length allows the model to capture relevant behavioral patterns while avoiding the dilution and noise associated with excessively long windows, thereby yielding comparatively better performance. Moreover, the class imbalance ratio in the test set is not linearly related to the optimal window length. Table 6 summarizes the selection of the length of the feature window based on the results obtained.

Fig. 4 illustrates the trends in the mean accuracy, recall, F1, and AUC over 20 independent trials as the feature window length varies, using dataset D6 as an example, highlighting the impact of feature window length on model performance. D6 is chosen as an example due to its pronounced relative ranking differences and substantial performance

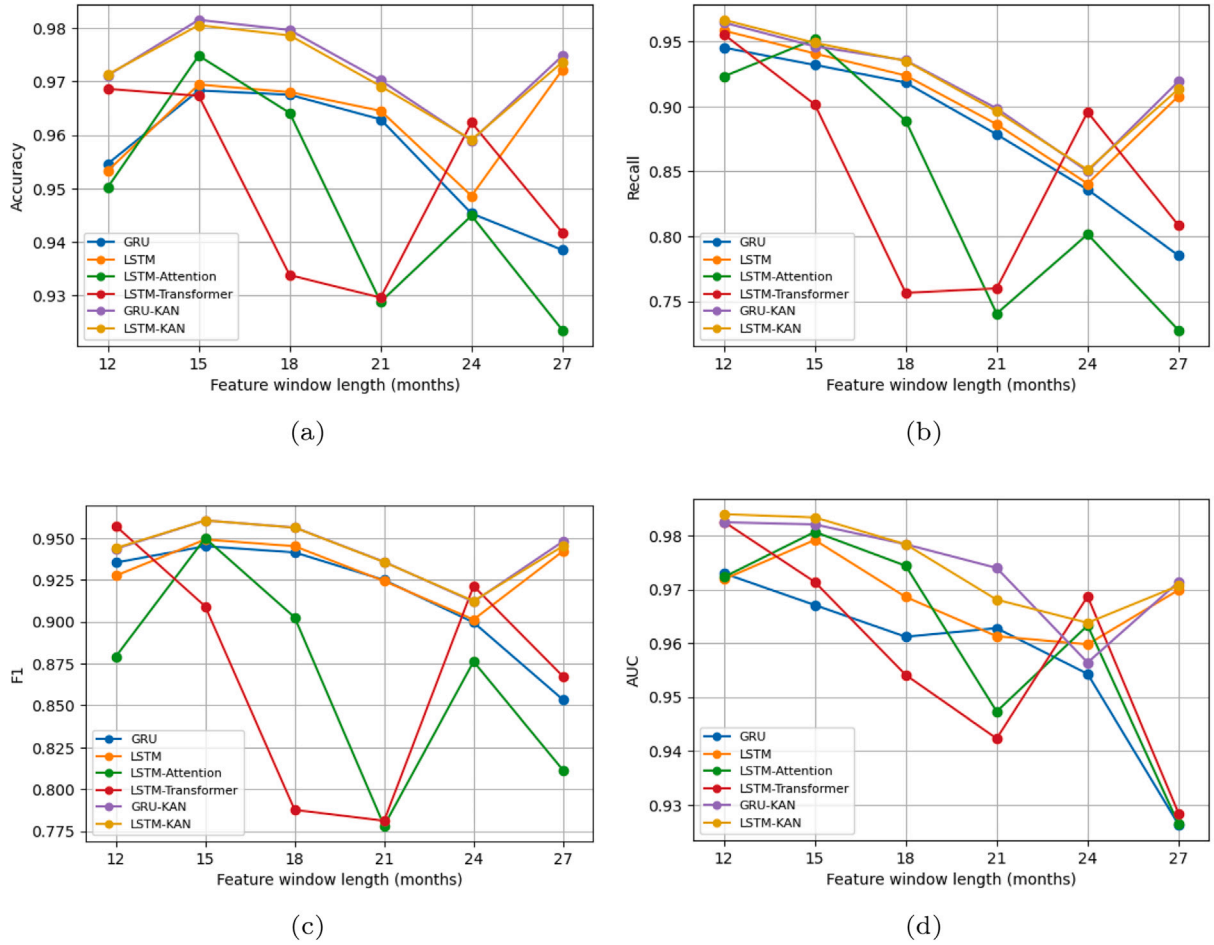


Fig. 4. Line charts depicting (a) accuracy, (b) recall, (c) F1 score, and (d) AUC for each model under different feature window lengths. The observation period is fixed at 3 months.

fluctuations, which facilitate illustrating the effects of varying feature window lengths. Detailed results, including the maximum values and standard deviations of all evaluation metrics over 20 independent trials, are provided in Table A2 in the appendix. Across the tested feature window lengths, the proposed GRU-KAN and LSTM-KAN models generally achieve high performance across most metrics. In addition, compared with the relatively high standard deviations observed in baseline models such as LSTM-Attention, the proposed models demonstrate superior stability.

GRU and LSTM show similar trends in accuracy, recall, and F1 scores, except for the longest window length, where LSTM's strong long-term dependency capture makes it more effective with longer time series than GRU. However, integrating GRU with the KAN layer greatly increases its performance by effectively modeling non-linear structures. The performance of GRU-KAN and LSTM-KAN is similar across various metrics, indicating that the KAN layer provides comparable optimization benefits, thus narrowing the feature learning differences of the models. In terms of AUC, the variations between GRU-KAN and LSTM-KAN mirror those between GRU and LSTM, as each performs better depending on the window length, though to different degrees.

Both LSTM-Attention and LSTM-Transformer exhibit significant variability across all metrics as the feature window length increases, indicating reduced robustness. The main difference between these models is the inclusion of a feedforward network in LSTM-Transformer, yet their performance trends are comparable. The difference lies in their window length preferences, with LSTM-Transformer better at managing long time series than LSTM-Attention. Specifically, these models perform

best with window lengths of 15 and 24, likely due to optimal alignment with the processing of the attention layer. This situation emphasizes the sensitivity of the models to specific feature window lengths.

Moreover, the effectiveness of Transformer-based models is closely tied to dataset characteristics. While Kim et al. [23] reported notable performance of Transformer-based models on the Yahoo S5 and NeurIPS-TS benchmarks, which diverges from the findings of this study, it is critical to recognize that the dataset employed in this study is a large-scale real-world financial dataset characterized by borrower heterogeneity, multi-dimensional features, and severe class imbalance. This contrasts with the univariate time series and synthetically structured data employed by Kim et al. [23]. This result is consistent with the findings of Porcelli et al. [41], which suggest that attention mechanisms may perform poorly in anomaly detection because they calculate the similarity between all pairs of time points, potentially being dominated by the prevalent normal patterns. In such complex situations, the advantages of Transformers in capturing temporal dependencies may be attenuated. Consequently, while Transformer-based models usually excel in processing synthetic or sensor-type series with long horizons and distinct attention patterns [42], their benefits may be less evident when applied to intricate and heterogeneous panel data.

Finally, GRU-KAN demonstrates superior overall performance in most cases. In scenarios without a blank interval, GRU may be more suitable than LSTM. This is because, in cases with relatively short sequences and less complex dependencies, GRU's gating mechanism effectively captures the necessary temporal relationships while avoiding the added complexity of LSTM's memory cells. However, as shown in Section 4.3,

where a blank interval is applied, LSTM-KAN retains an advantage in capturing temporal dependencies under long-term information loss (e.g., longer than 8 months), due to its explicit memory cells and more sophisticated gating mechanism, as noted by Furizal et al. [43] and Cahuantzi et al. [44]. The architectural differences between GRU and LSTM imply distinct trade-offs. Consequently, in practice, GRU-KAN is recommended for short- to medium-term early prediction tasks in financial time series, while LSTM-KAN may be more suitable for tasks involving long-term credit risk monitoring or customer behavior modeling.

Overall, the proposed models consistently showed better performance across different feature window lengths, validating their robustness and adaptability. This improvement in key metrics highlights their strength in capturing complex feature relationships and handling temporal data.

4.3. Early prediction performance and significance analysis

The early prediction condition is created by introducing a gap between x and y called blank interval, where x represents the window of input characteristics, and y is the observation period for prediction. In sections 4.1, Sections 4.2 and 4.4, the gap is set to zero, making x and y adjacent, which allows the model to use the latest features of the time period for prediction, eliminating any interval between the input and output periods.

In this subsection, blank interval levels are defined for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12 months to conduct a sensitivity analysis and evaluate the models' ability to predict defaults in advance. Here, the observation periods of 3 and 6 months are set as the target variable y , during which default may occur, corresponding to the evaluation of default prediction accuracy under short-to-medium and long observation horizons, respectively. The total duration of x and the blank interval remains at 21 months, altering the length of the input feature window with the interval. Fig. 5 shows the test set, in which the model predicts defaults for a duration from t to $t + 3$ months with data spanning $t - 21$ to $t - i$ months, where i months are denoted as the interval duration.

4.3.1. Early prediction performance analysis

Let B denote the set of blank intervals and $B = \{1, \dots, 12\}$. The definitions of D and M are the same as in Section 4.1. Let $recall_{d,m,b}$, $F1_{d,m,b}$, and $AUC_{d,m,b}$ represent the mean recall, F1, and AUC obtained from 20 independent trials when applying blank interval $b \in B$ and model $m \in M$ to dataset $d \in D$. Correspondingly, $recall_RANK_{B,M}(d, m, b)$, $F1_RANK_{B,M}(d, m, b)$, and $AUC_RANK_{B,M}(d, m, b)$ indicate their respective global rankings across all blank intervals and models within each dataset. The total average ranking for dataset d when using blank interval b and model m , denoted as $AvgR_{B,M}(d, m, b)$, is defined as:

$$AvgR_{B,M}(d, m, b) = \frac{AUC_RANK_{B,M}(d, m, b) + F1_RANK_{B,M}(d, m, b) + recall_RANK_{B,M}(d, m, b)}{3} \quad (22)$$

where $d \in D$, $b \in B$, and $m \in M$.

Subsequently, these values are grouped according to the models in $M = \{GRU, LSTM, LSTM\text{-}Attention, LSTM\text{-}Transformer, GRU\text{-}KAN, LSTM\text{-}KAN\}$, and their within-group averages are denoted as $AvgR_M$:

$$AvgR_M(d, m) = \frac{1}{|B(m)|} \sum_{b \in B(m)} AvgR_{B,M}(d, m, b) \quad (23)$$

where $m \in M$, and $B(m) \subset B$ denotes the group of blank intervals evaluated with model m .

Tables 7 and 8 present the $AvgR_M(d, m)$ when the observation periods are 3 and 6 months, respectively. $AvgR_M(d, m)$ reflects the relative performance of each model when considering the overall scenario across the 1–12 blank intervals. By comparing the two tables, it can be observed that as the observation window is extended from 3 months to 6 months, the relative performance differences among the models change: the ranking gap between GRU-KAN and LSTM-KAN narrows, the gap between LSTM-Attention and LSTM-Transformer also decreases, while the gap between GRU and LSTM widens. Regardless of whether the observation window is 3 months or 6 months, the default ratio in the test set affects the absolute performance of the six models but has only a minor impact on their relative rankings. In test sets with the original default ratio, the ranking gap between GRU-KAN and LSTM-KAN is smaller than that observed in test sets with balanced default ratios; conversely, the ranking gap between LSTM-Attention and LSTM-Transformer in the original default ratio test sets is larger than the corresponding gap in the balanced default ratio test sets. Furthermore, the performance of GRU relative to LSTM improves as the default ratio decreases, although the overall

Table 7

$AvgR_M(d, m)$ (lower is better) of the 20 datasets evaluated under different models when observation period y is 3 months.

Dataset	GRU	LSTM	LSTM-Attention	LSTM-Transformer	GRU-KAN	LSTM-KAN
D1	37.2	36.9	38.1	38.4	33.8	34.6
D2	36.8	37.2	37.9	38.3	33.9	34.9
D3	36.9	37.1	37.8	38.1	33.7	35.4
D4	37.1	37.2	37.5	38.4	34.0	34.7
D5	36.7	37.0	38.0	38.6	34.1	34.6
D6	37.2	36.8	37.8	38.3	34.1	34.8
D7	36.9	36.5	38.1	38.8	34.5	34.2
D8	37.2	36.5	36.8	38.4	34.5	35.6
D9	37.5	37.2	37.1	38.4	34.2	34.6
D10	36.2	36.9	37.9	38.7	34.5	34.8
D11	37.3	36.9	37.5	38.3	33.9	35.1
D12	36.7	37.1	37.8	38.4	34.1	34.9
D13	36.1	36.8	37.9	38.8	34.2	35.2
D14	36.5	36.9	37.8	38.6	34.4	34.8
D15	36.3	36.7	38.1	38.5	34.5	34.9
D16	36.9	37.0	37.3	37.8	34.3	35.7
D17	36.8	37.1	37.4	38.1	34.4	35.2
D18	36.7	37.1	37.3	37.9	34.4	35.6
D19	36.9	36.6	37.1	37.8	34.7	35.9
D20	36.4	36.9	37.3	37.6	34.9	35.8
Average	36.8	36.9	37.1	38.3	34.3	35.1

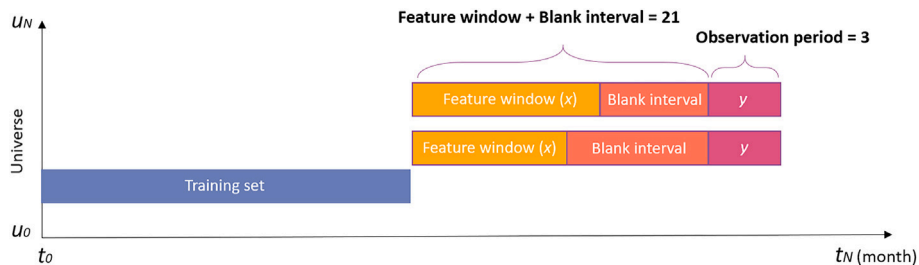


Fig. 5. Overview diagram of experiments using different blank interval configurations for different tests.

Table 8

$AvgR_M(d, m)$ (lower is better) of the 20 datasets evaluated under different models when observation period y is 6 months.

Dataset	GRU	LSTM	LSTM-Attention	LSTM-Transformer	GRU-KAN	LSTM-KAN
D1	37.1	36.8	37.8	38.1	34.3	34.9
D2	36.5	37.1	37.7	37.8	34.7	35.2
D3	36.7	37.0	37.5	37.7	34.9	35.2
D4	36.6	36.9	37.2	37.8	35.1	35.4
D5	36.1	36.8	37.9	38.2	34.9	35.1
D6	37.1	36.5	37.6	38.1	34.6	35.1
D7	37.4	36.8	37.5	37.9	34.8	34.6
D8	36.8	36.6	37.0	37.5	35.1	36.0
D9	36.9	36.7	37.2	37.7	35.3	35.2
D10	36.2	36.6	37.0	37.5	35.5	36.2
D11	37.0	36.4	37.3	37.9	34.8	35.6
D12	36.9	36.3	37.6	38.1	34.9	35.2
D13	36.8	36.9	37.3	37.7	34.8	35.5
D14	36.5	36.8	37.2	37.6	35.2	35.7
D15	35.7	36.5	37.8	38.1	35.3	35.6
D16	36.2	36.9	37.4	37.8	34.8	35.9
D17	36.4	36.9	37.2	37.8	35.2	35.5
D18	36.6	37.1	37.3	37.8	34.9	35.3
D19	36.7	37.0	37.9	38.2	34.7	34.5
D20	36.3	36.9	37.5	37.9	35.0	35.4
Average	36.6	36.8	37.5	37.8	35.0	35.4

trend is not strictly linear. Overall, our proposed models outperform the baselines across all test sets.

Although longer observation periods cover a broader range of default events, the predictive signals are less temporally precise. In contrast,

shorter periods yield predictions that are more actionable in practice, as they identify imminent risks with clearer timing, thereby enhancing the practical utility of the predictions. Therefore, Fig. 6 takes dataset D5 with a 3-month observation window as an example to visualize the mean evaluation metrics of the six models under different blank intervals, highlighting the overall declining trend as the blank interval increases. Dataset D5 was selected because it exhibits the largest relative ranking differences among all datasets with original default ratios. Since the blank interval is treated as a key input variable, varying it systematically allows for a sensitivity analysis, revealing how each model's performance responds to changes in temporal gaps and providing insights into model robustness under conditions of missing or sparse information. Detailed results, including the confusion matrix of a representative trial, the averaged accuracy, precision, recall, F1, and AUC over 20 independent runs, as well as the corresponding rankings of recall, F1, and AUC used to calculate $AvgR_M(d, m)$ in Table 7, are provided in Table A3 in the appendix.

The results show that as the blank interval increases, all six models exhibit varying degrees of decline in all evaluation metrics, though the magnitude of decline differs significantly. For recall, the models perform similarly in the initial intervals, generally remaining around 0.9, indicating that when the data gap is small, the models can still effectively identify useful information. However, as the interval widens, the curves of LSTM-Attention and LSTM-Transformer decrease sharply, reflecting insufficient generalization ability. In contrast, GRU-KAN and LSTM-KAN maintain higher levels with a slower decline, demonstrating the robustness of the KAN structure in scenarios with missing information, while the curves for GRU and LSTM consistently remain below those of

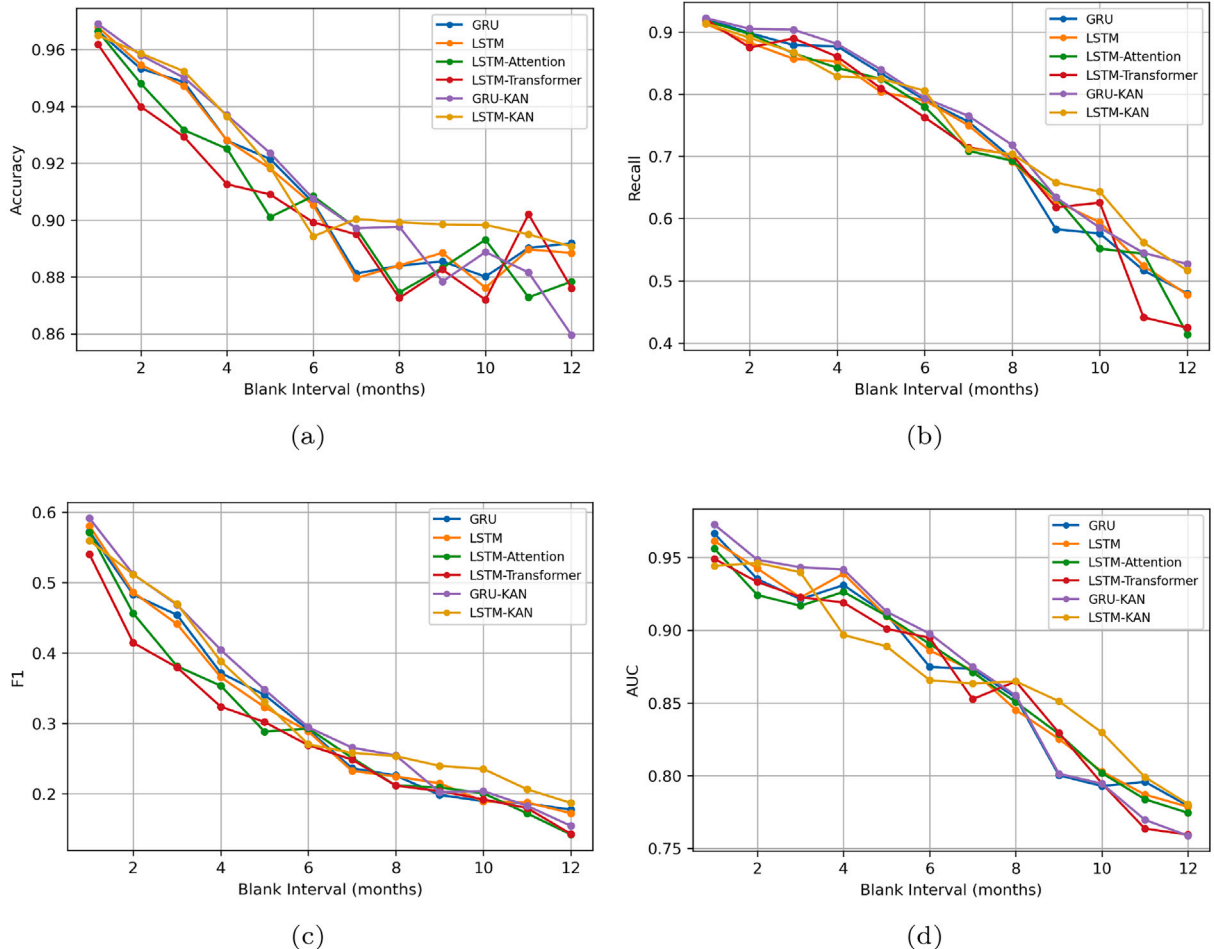


Fig. 6. Line charts displaying (a) accuracy, (b) recall, (c) F1 score, and (d) AUC for each model under different blank intervals.

the proposed models, showing no evident advantage. The overall trend of F1 is similar to that of recall but with a sharper decline, suggesting a severe imbalance between precision and recall. The decline in AUC is much smaller, as it is not calculated based on a fixed threshold and is therefore less sensitive to errors in individual predictions under class imbalanced conditions. In contrast, the reliability of accuracy as a performance metric diminishes under conditions of extreme class imbalance, as the large proportion of correctly predicted non-default users (TN) among all samples inflates its value. Moreover, the lack of smoothness in the accuracy curve reflects the nonlinear fluctuations in the number of non-default users misclassified as defaults (FP) with increasing blank interval, with the oscillations being most pronounced in the case of the LSTM-Transformer.

Across all metrics, GRU-KAN and LSTM-KAN consistently outperform the baselines, but their strengths appear in different ranges: GRU-KAN performs better for intervals of 1–8 months, while LSTM-KAN surpasses it in 9–12 months, consistent with the architectural differences discussed in Section 4.2. Notably, the proposed models achieve recall and AUC at blank interval t that are comparable to those of baseline models at $t - 1$, indicating that they can provide equally accurate default predictions one month earlier. This earlier warning window offers financial institutions a significant advantage in proactive risk management.

LSTM-Transformer exhibits the weakest performance among the baselines, suggesting that under early prediction settings, introducing blank intervals between the input feature window and the observation period creates significant challenges. Feature sparsity and information loss caused by longer gaps prevent the model from leveraging features close to the prediction target, while the complexity of self-attention imposes an additional computational burden on short or sparse sequences, weakening the inherent sequence modeling ability of LSTM. The decline in GRU can be attributed to the limitations of its gating mechanism. Although update and reset gates are designed to capture long-term dependencies, extended intervals lead to the loss of crucial temporal information, making it difficult to model complex dynamics over time. In contrast, the KAN layer focuses on learning non-linear relationships and adaptive activation functions, effectively extracting features related to default events even under long gaps or sparse conditions. This capability mitigates information loss and complements GRU and LSTM, thereby enhancing robustness and predictive performance.

4.3.2. Statistical significance analysis

Wilcoxon signed-rank tests are employed to evaluate whether there are statistically significant improvements in the evaluation metrics between the proposed models and the baseline models. This non-parametric test compares the ranks of the differences between paired observations, making it suitable for situations where the data are not normally distributed. A one-sided test is performed, with the null hypothesis stating that the proposed models exhibit equal or worse performance compared to the baseline models, while the alternative hypothesis posits that the proposed models perform better.

Following the experimental setup described in Section 4.3.1 and using the 3-month observation period, the Wilcoxon test was conducted based on the results of 20 independent trials for each dataset $d \in \mathcal{D}$ and blank interval length $b \in \mathcal{B}$ to assess the statistical significance of the key evaluation metrics, namely F1, AUC, and Recall. For each metric, the two proposed models (GRU-KAN and LSTM-KAN) were compared separately with four baseline models (GRU, LSTM, LSTM-Attention, and LSTM-Transformer), and their performance differences were also compared with each other. The model pairs and their abbreviations are listed in Table 9, comprising a total of 10 pairs. Consequently, for each dataset, blank interval configuration, and evaluation metric, 10 independent Wilcoxon tests were performed, yielding 10 corresponding p -values. This procedure was repeated across 20 datasets, 12 blank interval settings, and 3 evaluation metrics, resulting in 7200 Wilcoxon test outcomes in total. A small p -value indicates a statistically significant difference between the two models, with the significance level set at $\alpha = 0.01$.

Table 9

Model pairs and their abbreviations used in the Wilcoxon signed-rank test.

Model Pair (proposed model vs. baseline)	Abbreviation
GRU-KAN vs. GRU	WG1
GRU-KAN vs. LSTM	WG2
GRU-KAN vs. LSTM-Attention	WG3
GRU-KAN vs. LSTM-Transformer	WG4
LSTM-KAN vs. GRU	WL1
LSTM-KAN vs. LSTM	WL2
LSTM-KAN vs. LSTM-Attention	WL3
LSTM-KAN vs. LSTM-Transformer	WL4
GRU-KAN vs. LSTM-KAN	W5
LSTM-KAN vs. GRU-KAN	W6

However, conducting such a large number of statistical tests inevitably introduces the multiple comparisons problem [45], where the probability of Type I errors (false positives) accumulates across repeated hypothesis testing. Given the per-test significance level of $\alpha = 0.01$, approximately $7200 \times 0.01 = 72$ tests would be expected to appear significant purely by chance under the global null hypothesis.

To mitigate this issue, two standard procedures for controlling the false discovery rate (FDR) were applied: the Benjamini–Hochberg (BH) procedure [46] and the Benjamini–Yekutieli (BY) procedure [47]. The BH procedure controls the expected proportion of false discoveries among the rejected hypotheses, offering a less conservative alternative to family-wise error rate (FWER) control while maintaining high statistical power. The BY procedure generalizes BH by introducing a correction factor that ensures valid FDR control under arbitrary dependence among test statistics, making it suitable when correlations may exist across models, datasets, or evaluation metrics. In this study, all 7200 Wilcoxon signed-rank tests were treated as a single family of hypotheses, as they collectively assess the overall performance advantages of the proposed models across different datasets, blank-interval settings, and evaluation metrics. This global FDR control ensures a unified significance threshold across all tests, providing consistent statistical standards and enhancing the robustness of the global inference regarding model superiority, despite its conservativeness.

Given the large number of statistical tests conducted in this study, Table 10 summarizes the Wilcoxon signed-rank test results for each model pair under different significance criteria, reporting both the counts and ratios of significant outcomes. To reflect the robustness of the results under varying levels of statistical stringency, the table presents the test outcomes based on three significance determination standards. First, the unadjusted significance level is set to $\alpha = 0.01$. Second, under the target false discovery rate (FDR) level of $q = 0.01$, the Benjamini–Hochberg (BH-FDR) procedure was applied to all p -values to determine the critical threshold, resulting in an adjusted significance level of $\alpha_{BH} = 5.34 \times 10^{-3}$. Finally, in cases where dependency among test statistics may exist, the Benjamini–Yekutieli (BY-FDR) procedure introduces an additional dependency correction based on the BH-FDR method, yielding a more stringent threshold of $\alpha_{BY} = 5.08 \times 10^{-4}$, thereby ensuring effective control of the false discovery rate at the target level of $q = 0.01$ even under correlated conditions. Based on all 20 datasets and 12 blank interval configurations, each model pair was subjected to a total of 240 Wilcoxon tests. To provide a clearer comparison of model performance across different prediction horizons, and following the experimental conclusions in Section 4.3.1, the results for blank intervals of 1–8 months and 9–12 months are presented separately. For each model pair, the ratio for the 1–8 month intervals was calculated as the number of significant results divided by 160, while the ratio for the 9–12 month intervals was calculated as the number of significant results divided by 80. This design facilitates a more intuitive comparison between GRU-KAN and LSTM-KAN, highlighting their respective statistical advantages in short- to medium-term and long-term prediction tasks.

Table 10
Wilcoxon significance results under different significance levels for each model pair and metric.

metric	pair	Interval 1 to 8						Interval 9 to 12					
		$\alpha^{(a)}$		$\alpha_{BH}^{(b)}$		$\alpha_{BY}^{(c)}$		$\alpha^{(a)}$		$\alpha_{BH}^{(b)}$		$\alpha_{BY}^{(c)}$	
		count	ratio	count	ratio	count	ratio	count	ratio	count	ratio	count	ratio
F1	WG1	142	89%	117	73%	106	66%	32	40%	26	33%	21	26%
	WG2	144	90%	121	76%	114	71%	31	39%	19	24%	12	15%
	WG3	148	93%	128	80%	121	76%	45	56%	23	29%	10	13%
	WG4	152	95%	147	92%	134	84%	44	55%	23	29%	11	14%
	WL1	104	65%	96	60%	83	52%	62	78%	57	71%	57	71%
	WL2	118	74%	105	66%	97	61%	72	90%	58	73%	46	58%
	WL3	125	78%	103	64%	93	58%	77	96%	72	90%	72	90%
	WL4	149	93%	129	81%	116	73%	78	98%	76	95%	76	95%
AUC	W5	81	51%	57	36%	32	20%	8	10%	3	4%	0	0%
	W6	16	10%	4	3%	0	0%	71	89%	66	83%	62	78%
	WG1	112	70%	107	67%	95	59%	14	18%	11	14%	5	6%
	WG2	147	92%	125	78%	116	73%	13	16%	10	13%	7	9%
	WG3	141	88%	122	76%	113	71%	15	19%	11	14%	7	9%
	WG4	149	93%	132	83%	127	79%	32	40%	28	35%	11	14%
	WL1	83	52%	70	44%	61	38%	62	78%	57	71%	44	55%
	WL2	95	59%	81	51%	72	45%	64	80%	64	80%	48	60%
Recall	WL3	98	61%	86	54%	83	52%	75	94%	68	85%	64	80%
	WL4	108	68%	94	59%	88	55%	78	98%	76	95%	76	95%
	W5	95	59%	89	56%	64	40%	6	8%	5	6%	0	0%
	W6	16	10%	15	9%	12	8%	72	90%	70	88%	62	78%
	WG1	115	72%	98	61%	91	57%	68	85%	60	75%	57	71%
	WG2	145	91%	118	74%	112	70%	47	59%	41	51%	40	50%
	WG3	141	88%	129	81%	116	73%	45	56%	43	54%	38	48%
	WG4	154	96%	149	93%	123	77%	59	74%	47	59%	41	51%
	WL1	106	66%	90	56%	83	52%	79	99%	56	70%	52	65%
	WL2	104	65%	92	58%	76	48%	71	89%	63	79%	61	76%
	WL3	91	57%	78	49%	64	40%	77	96%	70	88%	66	83%
	WL4	99	62%	82	51%	77	48%	78	98%	74	93%	70	88%
	W5	82	51%	71	44%	46	29%	25	31%	6	8%	2	3%
	W6	21	13%	16	10%	0	0%	51	64%	43	54%	41	51%

a Unadjusted critical significance level $\alpha = 0.01$. b Critical significance level after BH-FDR correction $\alpha_{BH} = 5.34 \times 10^{-3}$.
c Critical significance level after BY-FDR correction $\alpha_{BY} = 5.08 \times 10^{-4}$.

From Table 10, it can be observed that within the 1–8 month blank interval, the proposed GRU-KAN model exhibits a high proportion of significant results when compared with all baseline models (WG1–WG4). For most relevant model pairs, the significance rate exceeds 90% under the unadjusted threshold and remains above 70% even after applying the BH-FDR and BY-FDR corrections. In contrast, under longer blank intervals (9–12 months), the LSTM-KAN model (WL1–WL4) demonstrates stronger dominance. The direct comparisons between the two proposed models (W5 and W6) further reveal their complementary characteristics: GRU-KAN performs significantly better than LSTM-KAN in short-term prediction settings, whereas the opposite holds true in long-term prediction scenarios exceeding 8 months. It is worth noting that some results that were significant under the unadjusted threshold became non-significant after FDR correction, indicating that their statistical evidence was not sufficiently strong to remain significant under stricter FDR control; however, this does not imply that they are false positives. Table A4 in the Appendix provides the detailed original p -values and their significance annotations for each blank interval setting using dataset $D5$ as an example. Overall, most model pairs retain significance after applying BH-FDR and BY-FDR corrections, suggesting that the observed performance differences are statistically robust. These findings collectively confirm the complementary advantages of the two proposed models across different early prediction horizons.

4.4. Sample size performance analysis

Various sample size configurations were employed to evaluate the feasibility and predictive capability of the proposed models under limited data conditions. The total duration of the training and testing window was set at 18 months, with the feature window of 15 months and the observation period of 3 months, following the optimal length

Table 11
Configurations overview for various sample sizes for training and test sets.

Sample size	Training set (2019)			Test set (2020)		
	No. of Eligible Loans	No. of Default	Default Rate (%)	No. of Eligible Loans	No. of Default	Default Rate (%)
500,000	10,906	249	2.28	10,777	149	1.38
1,000,000	21,902	464	2.12	21,521	300	1.39
1,500,000	33,008	706	2.14	32,292	432	1.34
2,000,000	44,135	934	2.12	43,134	584	1.35
3,000,000	66,427	1373	2.07	64,782	862	1.33
5,000,000	110,946	2271	2.05	108,086	1476	1.37

obtained in Section 4.2. No blank interval is utilized as early prediction is not included here. Sequences less than 18 months (15+3 months) were excluded due to inadequate historical data for effective training. Longer sequences were truncated at 18 months for consistency. Table 11 presents the number of loans with a minimum duration of 18 months, grouped by loan sequence number, as well as the number of defaulted loans and the corresponding default rates across six repayment record sample sizes (500,000, 1,000,000, 1,500,000, 2,000,000, 3,000,000, and 5,000,000 entries), using 2019 data as the training set and 2020 data as the test set.

The models were trained and tested on a machine equipped with an 11th Gen Intel Core i7-11375H CPU, 16 GB RAM, Intel Iris Xe Graphics, and an NVIDIA GeForce RTX 3060 Laptop GPU. Table 12 reports the training time per epoch for the six models using the highest sample size in Table 11. Since class imbalance is not the focus of this subsection, the training and test sets were processed with 1:1 random undersampling to address the issue, resulting in an actual training sample size equal to twice that of the minority class, which is 4542 time series.

Table 12

Training time per epoch for the six models with the highest sample size in Table 11.

Model	Training time (s)	Relative cost (vs. GRU)
GRU	21.6673	1.0×
LSTM	27.4397	1.27×
LSTM-Attention	25.7071	1.19×
LSTM-Transformer	261.6358	12.10×
GRU-KAN	29.2883	1.35×
LSTM-KAN	31.4780	1.45×

The training time results in Table 12 highlight the differences in computational cost among the six models. GRU achieves the fastest training, while LSTM incurs a slightly longer training time due to additional gating operations and memory cell computations. The addition of Attention and KAN layers further increases complexity, with GRU-KAN and LSTM-KAN requiring approximately 1.35 and 1.45 times the training time of GRU, respectively. LSTM-Transformer exhibits the highest cost, more than 12 times that of GRU, reflecting the substantial overhead of multi-head self-attention and feed-forward layers in sequential processing. To ensure fairness, all models were tuned following the same protocol, with iterative adjustments until the validation performance plateaued and early stopping was applied to prevent overfitting. Although the LSTM-Transformer model required considerably higher computational cost, its tuning process was not curtailed due to resource

limitations, as the available computational resources under the selected data scale were sufficient to support fair and thorough experimentation. These measurements provide a valuable complement to predictive performance, thereby highlighting the trade-off between accuracy and efficiency, and offering guidance for model selection under resource constraints. Overall, GRU-KAN achieves a favorable balance between predictive accuracy and computational cost.

Fig. 7 displays line charts based on the average results of 20 independent trials, showing the performance trends across varying sample sizes. GRU-KAN and LSTM-KAN demonstrate comparable performance, alternating in superiority across different sample size configurations. LSTM emerges as the highest performing baseline, closely trailing GRU-KAN and LSTM-KAN in all metrics, indicating that KAN improves GRU more than LSTM. The curves of GRU-KAN, LSTM-KAN, and LSTM exhibit stable trends with low fluctuations. In contrast, GRU, LSTM-Attention, and LSTM-Transformer perform poorly with small samples, suggesting that they require larger datasets for reliable performance. Detailed numerical results are provided in Table A5, which summarizes the mean, best, and standard deviation of 20 independent trials of accuracy, precision, recall, F1 score, and AUC. The findings confirm that performance gaps between models are more noticeable with smaller sample sizes but diminish as the sample size grows. Both proposed models consistently outperform the baseline models, displaying improved performance and stability across sample sizes, illustrating their effectiveness with limited data.

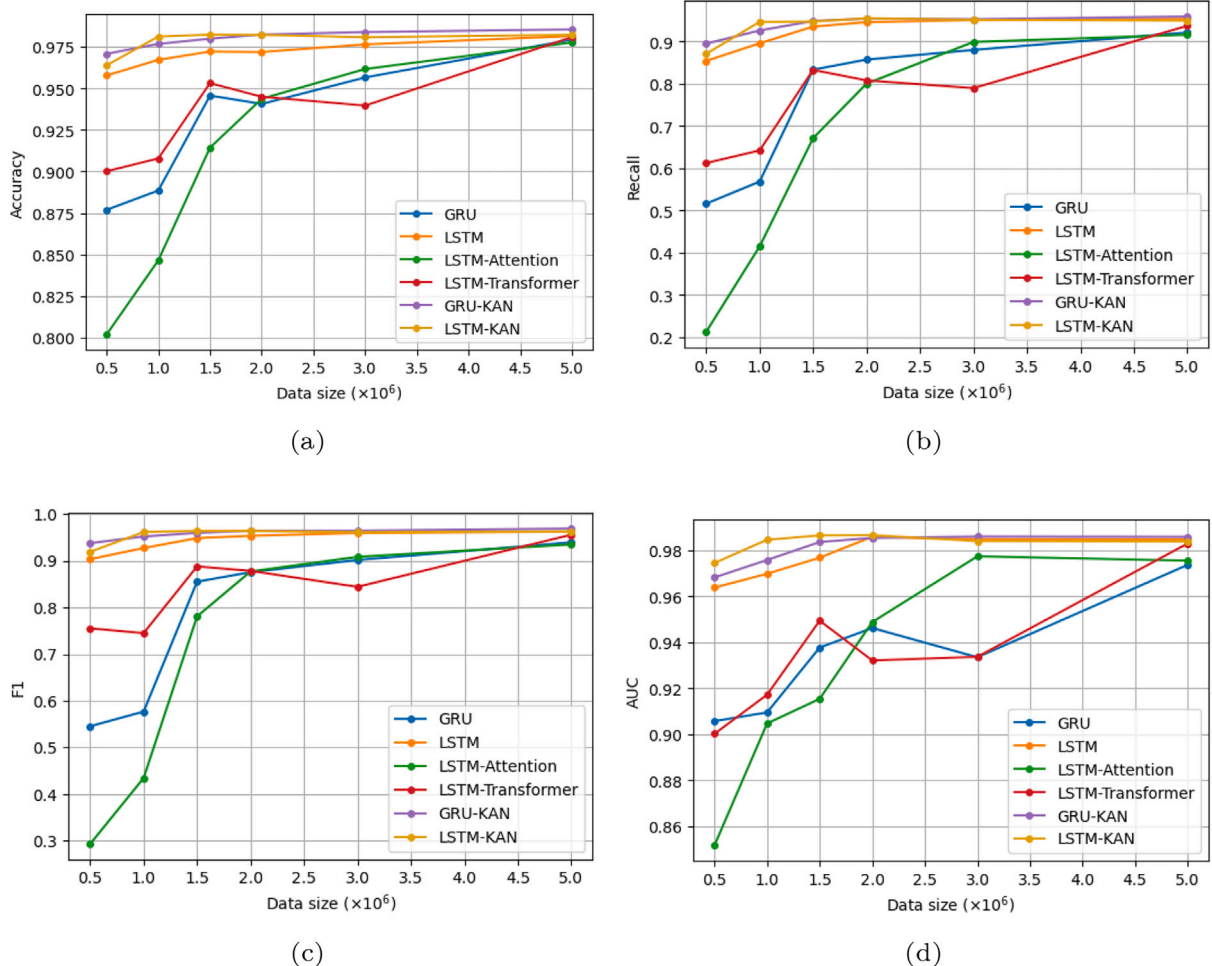


Fig. 7. Line charts displaying (a) accuracy, (b) recall, (c) F1 score, and (d) AUC for each model in different sample sizes.

Although these results highlight the advantages of the proposed architectures over conventional sequential models, it is important to contextualize them alongside survival analysis approaches [4], which remain fundamental in credit risk management for long-term forecasting. Survival analysis uses complete lifecycle data from closed loans to model the entire credit cycle from origination, making it well suited for long-term strategic assessments. In contrast, our model learns borrower behavioral patterns from shorter loan sequence segments without requiring full historical records, making it more applicable in contexts with limited data availability or newly issued products lacking complete lifecycle information. Together, these methodologies are highly complementary within modern credit risk frameworks in which survival analysis provides robust long-term estimation, while our approach offers flexible, data-efficient short-term early prediction.

4.5. Model generalization performance analysis

To evaluate the generalization ability of the proposed models on the original default rate data, experiments were conducted using cohorts from different years, assessing their robustness against concept drift, a phenomenon widely recognized in time series prediction tasks, where data distributions evolve over time. When trained on data covering an entire year, the models were exposed to fluctuations across multiple quarters. This suggests that the models may have captured typical annual patterns in an aggregated sense, but not necessarily short-term quarterly variations. To address this limitation, additional cross-quarter experiments were carried out using cohorts from different quarters.

Each annual or quarterly cohort consists of monthly repayment data from loan applicants originating in a specific quarter or year, forming an independent dataset. For the multi-year (covering two or more years) experiments, six training–testing pairs were constructed in total. Specifically, the 2018 dataset was used for training, with testing conducted on 2020–2022 data (D_{10} , D_{15} , D_{20}); the 2019 dataset (D_5) was used for training, with testing on 2021 and 2022 data (D_{15} , D_{20}); and the 2020 dataset (D_{10}) was used for training, with testing on 2022 data (D_{20}). Data from 2023 onward were excluded, as they were not yet fully updated in the existing dataset.

For the cross-quarter experiments, the 2018 Q1 dataset was used for training, with testing on 2018 Q2, 2018 Q3, 2018 Q4, and 2019 Q1. Similarly, the 2018 Q2 dataset was used for training, with testing on 2018 Q3, 2018 Q4, 2019 Q1, and 2019 Q2, resulting in a total of eight training–testing pairs. The feature window length was set to 15 months, with a blank interval of 3 months and an observation period of 3 months in all training–testing pairs. Table 13 presents the statistics for each cohort in the multi-year and cross-quarter experiments, including the number of unique loans, the number of loans meeting the 21-month data length criterion ($15 + 3 + 3$), the number of defaults, and the corresponding default rates.

Table 13

Summary of total loan configurations and the count of loans meeting the 21-month eligibility criterion under different cohorts.

Cohort	No. of Unique Loans	No. of Eligible Loans	No. of Defaults	Default Rate (%)
2018	123,160	82,914	1543	1.86
2019	171,609	94,689	2735	2.89
2020	381,022	298,939	1174	0.39
2021	396,041	361,275	1195	0.33
2022	150,882	138,270	953	0.69
2018 Q1	28,474	22,853	187	0.82
2018 Q2	34,955	24,921	245	0.98
2018 Q3	32,305	20,417	442	2.16
2018 Q4	27,426	14,723	669	4.54
2019 Q1	26,787	13,945	616	4.42
2019 Q2	39,781	20,757	769	3.70

Tables A6 and A7 in the appendix summarize the average, best, and standard deviation of model performance over 20 independent trials in the multi-year and cross-quarter experiments, while Fig. 8 shows bar charts of the average F1 and AUC. Performance variation closely tracks the magnitude of the distributional change between training and testing periods. For example, predicting 2022 loan defaults using 2018 data produces a significant decline across all metrics relative to using 2019 data, highlighting a strong concept drift. Under such conditions, overall accuracy declines sharply and inter-model disparities widen, with GRU-KAN and LSTM-KAN demonstrating the greatest resilience. However, performance degradation does not increase monotonically with temporal distance, suggesting that the effects of concept drift depend more than just on the time gap.

Compared to the multi-year setting, the cross-quarter experiments, which are characterized by smaller sample sizes and the absence of annual patterns, exhibit greater variability of performance. However, the proposed models consistently outperform the baselines, with GRU-KAN achieving higher F1 and AUC in four of the six multi-year groups and six of the eight cross-quarter groups, demonstrating robust adaptability and generalization under changing temporal distributions. Although other models occasionally perform better in particular cohorts, for example, LSTM-Attention trained on 2018 Q2 data outperforms configurations using other periods, this finding highlights that predictive effectiveness depends not only on temporal proximity, but also on the representativeness and information richness of the training cohort and the model's ability to capture important patterns.

4.6. Ablation study

To evaluate the contribution of each component in the proposed models, an ablation study was conducted. For GRU-KAN, four variants were examined: the full model, GRU-MLP (M1), GRU only (M2), and KAN only (M3). Similarly, LSTM-KAN was compared with LSTM-MLP (M1'), LSTM only (M2'), and KAN only (M3'). Given that earlier results consistently demonstrated stable relative rankings of the six models across different years, the ablation analysis was restricted to D_1 (2018–2019, balanced ratio) and D_5 (2018–2019, original ratio). This configuration captures both balanced and highly imbalanced conditions, thereby ensuring representativeness while avoiding redundancy. Thus, the ablation serves its auxiliary role of assessing component contributions rather than reiterating the stability of model comparisons. The performance metrics included Accuracy, Precision, Recall, F1, and AUC. The input window length was fixed at 12, with a blank interval and observation period both set to 3. Although a window length of 12 is not optimal for datasets D_1 and D_5 as shown in Table 6, it was selected to avoid repetition of the 15 and 18 windows discussed in the previous subsection, and to assess the model's performance with a non-optimal window length.

Tables 14 and Table 15 summarize the average evaluation metrics of the ablation study based on 20 trials for GRU-KAN and LSTM-KAN, respectively. The results show that the full models consistently outperform their ablated variants, indicating a complementary relationship between GRU or LSTM and the KAN layer. The relative performance of these models exhibits different trends on the balanced dataset (D_1) and the dataset with the original default ratio (D_5), revealing the impact of class imbalance on model performance. Specifically, in the highly imbalanced D_5 , where non-default users dominate, models achieve higher accuracy but experience substantial declines in precision and F1, while recall remains largely unaffected. This suggests that in real-world scenarios, models are easily biased by the majority class, thereby overrating the accuracy. In this context, the full models GRU-KAN and LSTM-KAN demonstrate a pronounced advantage in precision compared to variants where the KAN layer is replaced with alternative components (M1 and M1') or removed (M2 and M2'), highlighting the critical role of KAN in improving the precision of positive class identification. Although the KAN-only variants (M3) achieve precision levels comparable to the full models, they

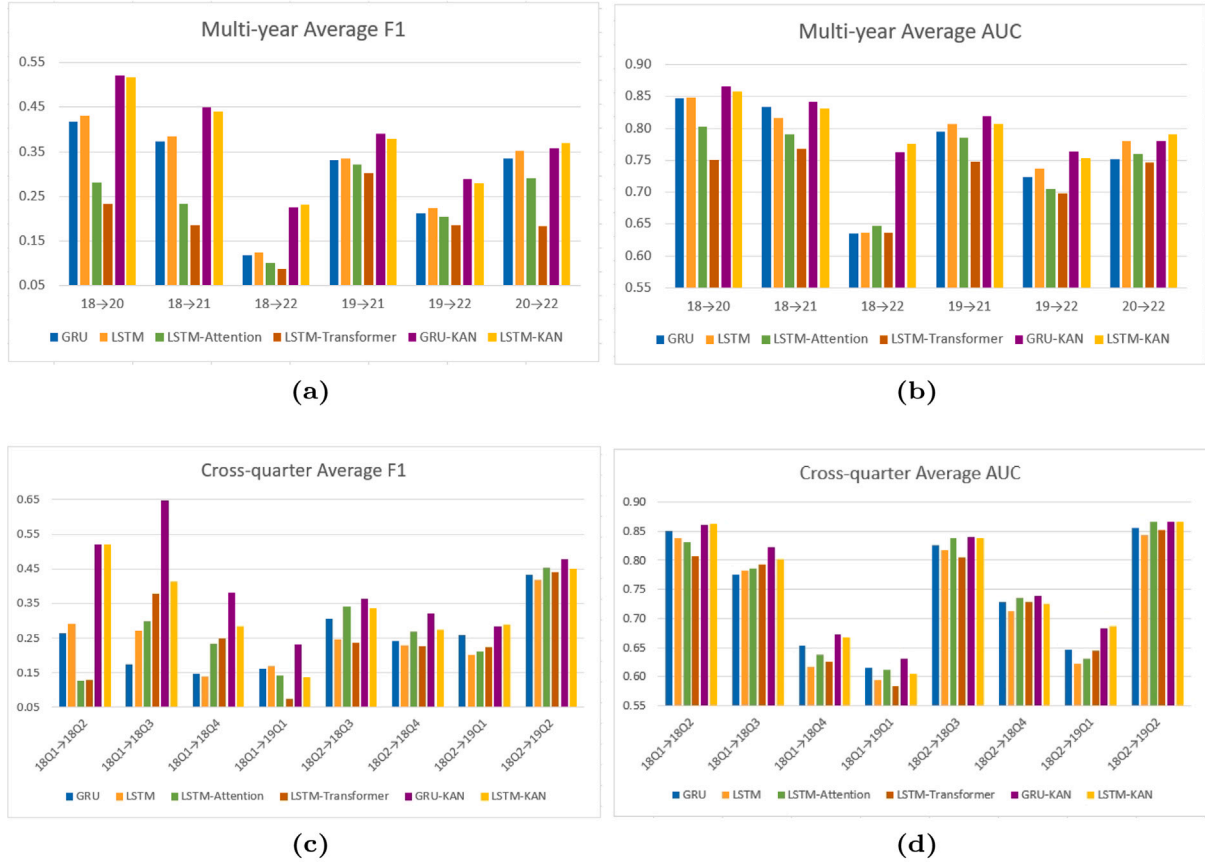


Fig. 8. Bar charts displaying (a) average F1 score of multi-year experiment, (b) average AUC of multi-year experiment, (c) average F1 score of cross-quarter experiment, and (d) average AUC of cross-quarter experiment. The number preceding the “→” signifies the training data year/quarter, while the number following it denotes the test data year/quarter.

Table 14

Ablation study on GRU-KAN across *D1* and *D5* datasets (feature window = 12, blank interval = 3, observation period = 3).

Dataset	Metric	GRU-KAN (Full)	GRU-MLP (M1)	GRU only (M2)	KAN only (M3)
D1	Accuracy	0.8941	0.8693	0.8703	0.8611
	Precision	0.9261	0.9214	0.9154	0.9150
	Recall	0.8567	0.8075	0.8159	0.7963
	F1	0.8900	0.8607	0.8628	0.8515
	AUC	0.9210	0.9067	0.8987	0.8926
D5	Accuracy	0.9318	0.9197	0.9189	0.9282
	Precision	0.2681	0.2321	0.2324	0.2515
	Recall	0.8356	0.8145	0.8281	0.7958
	F1	0.4059	0.3613	0.3629	0.3822
	AUC	0.9207	0.9168	0.9127	0.8980

Table 15

Ablation study on LSTM-KAN across *D1* and *D5* datasets (feature window = 12, blank interval = 3, observation period = 3).

Dataset	Metric	LSTM-KAN (Full)	LSTM-MLP (M1')	LSTM only (M2')	KAN only (M3)
D1	Accuracy	0.8796	0.8705	0.8621	0.8611
	Precision	0.9224	0.9185	0.8928	0.9150
	Recall	0.8290	0.8131	0.8230	0.7963
	F1	0.8732	0.8626	0.8564	0.8515
	AUC	0.9105	0.9092	0.9078	0.8926
D5	Accuracy	0.9284	0.9134	0.9171	0.9282
	Precision	0.2564	0.2190	0.2277	0.2515
	Recall	0.8248	0.8192	0.8239	0.7958
	F1	0.3912	0.3455	0.3568	0.3822
	AUC	0.9193	0.9172	0.9146	0.8980

exhibit substantial deficiencies in recall, underscoring the necessity of integrating KAN with recurrent structures (GRU or LSTM) to balance precision and recall.

A comparison across datasets further reveals that the superiority of GRU-KAN and LSTM-KAN over M1 and M1' is more pronounced in the imbalanced scenario (*D5*), indicating that KAN is more effective than MLP in handling imbalanced data, whereas their advantage over M2, M2', and M3 is more evident in the balanced scenario (*D1*). M2 and M2' exhibit higher recall compared to M1 and M1', which leads to nonlinear variations in F1. This phenomenon indicates that in both balanced and imbalanced datasets, recurrent structures (GRU or LSTM) contribute

more significantly to enhancing recall, whereas the KAN layer is essential for maintaining precision and stabilizing overall performance. Moreover, in *D1*, model performance follows a more linear trend. For example, precision and AUC decline smoothly from GRU-KAN to M3, and accuracy, F1, and AUC exhibit a similarly linear decrease from LSTM-KAN to M3.

In addition, the results demonstrate that GRU-KAN outperforms LSTM-KAN across most metrics. One possible explanation lies in their architectural differences: GRU employs a simpler gating mechanism with fewer parameters and omits the separate memory cell used in LSTM. This lighter design facilitates more efficient gradient propagation and makes GRU particularly well-suited for capturing short- and medium-term

dependencies that are common in financial time series. When combined with KAN, which excels at modeling nonlinear relationships and refining decision boundaries in imbalanced data, GRU's efficiency allows the complementary strengths of the two modules to be better realized. In contrast, the more complex LSTM structure may dilute this synergy, resulting in comparatively lower performance.

In summary, these findings suggest that in real-world loan default prediction, KAN effectively captures structural patterns, supporting improved precision in practice, while GRU and LSTM preserve temporal dependencies, which can contribute to enhanced recall. Integrating the two can enhance overall predictive performance and improve robustness across datasets with varying class distributions.

5. Limitations and future works

This study is limited to the Freddie Mac dataset. Although it is large and representative of real-world loan data, its scope is constrained to a single U.S. mortgage market and may reflect specific geographic or demographic patterns that limit broader applicability. The lack of access to other proprietary or industry datasets also prevented wider validation. Future work should incorporate additional datasets from different financial institutions and regions to better assess model robustness and generalizability across diverse financial and demographic contexts. Moreover, future research may focus on developing adaptive models capable of addressing concept drift, enabling dynamic adjustment to evolving data distributions for more reliable and accurate predictions in environments subject to substantial temporal changes.

6. Conclusion

The GRU-KAN and LSTM-KAN models proposed in this study enhance proactive risk management in loan default prediction by enabling reliable early default predictions months in advance. The study also addresses the gap in related research by using the LSTM-Transformer as a benchmark model to evaluate the performance of a Transformer-based model in loan default prediction. Research based on 20 benchmark datasets indicates that RUS is the most suitable resampling method for the datasets used in this study among the six models on average, although this does not mean it is optimal for every individual model. The optimal feature window length varies by dataset, with 12–18 months generally being the most suitable for the benchmark datasets in this study. To simulate early prediction scenarios, this study introduces the novel concept of the “blank interval” and conducts a sensitivity analysis for blank intervals ranging from 1 to 12 months. The results show that the length of the blank interval is approximately linearly negatively correlated with various model evaluation metrics. The proposed GRU-KAN and LSTM-KAN models outperform other baseline models across different blank interval lengths, with GRU-KAN being more advantageous in the 1–8 month blank interval and LSTM-KAN excelling in the 9–12 month interval. Additionally, the proposed models deliver more accurate predictions with smaller sample sizes and consistently demonstrate superior performance and robustness in the presence of concept drift. These characteristics are critical in the financial sector, where timely and accurate predictions are decisive for risk control and decision-making. Finally, the ablation study validates the independent contributions of each component, showing that the recurrent structure

(GRU or LSTM) significantly improves recall, while the KAN layer has a greater impact on precision. Thus, the combination of KAN and the recurrent structure complements each other, and their integration is crucial for maintaining the balance between precision and recall.

From a practical perspective, the two proposed models each exhibit distinct strengths, offering clear guidance for deployment in financial applications. The proposed models exhibit stable and robust performance under various data configurations, demonstrating their adaptability to dynamic financial environments where class imbalance is frequent and feature window settings may need to be fine-tuned for different applications. Specifically, GRU-KAN is more suitable for short- to medium-term early-prediction tasks. For instance, in short-term consumer loan management, financial institutions often need to issue default warnings several months in advance to initiate timely intervention measures. In such scenarios, GRU-KAN's lightweight gating structure ensures fast convergence and high responsiveness while maintaining competitive accuracy. In contrast, LSTM-KAN is better suited for long-term strategic monitoring and risk assessment, such as long-term mortgage risk forecasting or post-loan portfolio surveillance. Its explicit memory-cell design enables the model to preserve long-term temporal dependencies and more accurately capture the gradual evolution of borrower creditworthiness.

In conclusion, the GRU-KAN and LSTM-KAN models proposed in this study significantly enhance loan default prediction by effectively capturing temporal information. These models demonstrate higher predictive accuracy and efficiency in out-of-time evaluation compared to existing baseline models. This breakthrough enables the effective utilization of vast historical data, and their ability to predict defaults months in advance makes them highly practical for complex financial applications.

CRediT authorship contribution statement

Yue Yang: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zihan Su:** Writing – original draft, Validation, Software, Methodology. **Ying Zhang:** Writing – original draft, Validation, Software, Methodology. **Chang Chuan Goh:** Visualization, Software. **Yuxiang Lin:** Writing – original draft, Validation. **Anthony Graham Bellotti:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Boon Giin Lee:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Anthony Graham Bellotti reports that financial support was provided by Ningbo Science and Technology Bureau. Anthony Graham Bellotti reports a relationship with Ningbo Science and Technology Bureau that includes funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1
Hyperparameter search spaces and best configurations for baseline models.

Model	Parameter	Search Space/values	Best configuration
LSTM	Hidden size	32, 64, 128, 256, 512	128 (layer 1), 64 (layer 2)
	Num layers	1, 2, 3, 4, 5	2
	Dropout	0, 0.1, 0.2, 0.3, 0.4, 0.5	0.2
	Learning rate	1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3	1e-3
	Optimizer	Adam, AdamW	Adam
GRU	Hidden size	32, 64, 128, 256, 512	128 (layer 1), 64 (layer 2)
	Num layers	1, 2, 3, 4, 5	2
	Dropout	0, 0.1, 0.2, 0.3, 0.4, 0.5	0.1
	Learning rate	1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3	1e-3
	Optimizer	Adam, AdamW	Adam
LSTM-Attention	LSTM hidden size	32, 64, 128, 256, 512	128 (layer 1), 64 (layer 2)
	LSTM num layers	1, 2, 3, 4, 5	2
	LSTM dropout	0, 0.1, 0.2, 0.3, 0.4, 0.5	0.2
	Attention hidden size	32, 64, 128	64
	Attention dropout	0, 0.1, 0.2, 0.3	0.1
	Learning rate	1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3	1e-3
	Optimizer	Adam, AdamW	Adam
LSTM-Transformer	LSTM hidden size	32, 64, 128, 256, 512	128 (layer 1), 64 (layer 2)
	LSTM num layers	1, 2, 3, 4, 5	2
	LSTM dropout	0, 0.1, 0.2	0.1
	Transformer d_model	32, 64, 128, 256, 512	256
	Transformer num layers	1, 2, 3, 4, 5	1
	Transformer num heads	2, 4, 8	4
	FFN expansion ratio	2, 4	2
	Transformer dropout	0, 0.1, 0.2, 0.3, 0.4, 0.5	0.2
	Learning rate	1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3	1e-3
	Optimizer	Adam, AdamW	Adam

Table A2

Results of model performance on the *D6* dataset based on 20 trials using different input feature window lengths.

Window length	Model	Accuracy (Best, Std)	Precision (Best, Std)	Recall (Best, Std)	F1 (Best, Std)	AUC (Best, Std)
12	GRU	0.9546 (0.9594, 0.0021)	0.9256 (0.9479, 0.0118)	0.9453 (0.9678, 0.0074)	0.9353 (0.9426, 0.0034)	0.9730 (0.9769, 0.0031)
	LSTM	0.9533 (0.9587, 0.0026)	0.8985 (0.9174, 0.0099)	0.9584 (0.9614, 0.0017)	0.9275 (0.9309, 0.0047)	0.9719 (0.9834, 0.0009)
	LSTM-Attention	0.9502 (0.9679, 0.0472)	0.8391 (0.9121, 0.1979)	0.9231 (0.9751, 0.2173)	0.8791 (0.9377, 0.2070)	0.9723 (0.9803, 0.0267)
	LSTM-Transformer	0.9686 (0.9721, 0.0024)	0.9489 (0.9545, 0.0044)	0.9358 (0.9473, 0.0090)	0.9423 (0.9539, 0.0042)	0.9725 (0.9816, 0.0014)
	GRU-KAN	0.9711 (0.9764, 0.0039)	0.9235 (0.9430, 0.0147)	0.9646 (0.9682, 0.0036)	0.9435 (0.9534, 0.0071)	0.9825 (0.9854, 0.0039)
	LSTM-KAN	0.9713 (0.9731, 0.0012)	0.9224 (0.9283, 0.0048)	0.9667 (0.9700, 0.0017)	0.9440 (0.9473, 0.0023)	0.9840 (0.9843, 0.0003)
15	GRU	0.9683 (0.9742, 0.0010)	0.9586 (0.9626, 0.0049)	0.9320 (0.9412, 0.0069)	0.9451 (0.9565, 0.0061)	0.9671 (0.9743, 0.0082)
	LSTM	0.9694 (0.9805, 0.0014)	0.9575 (0.9696, 0.0073)	0.9408 (0.9544, 0.0036)	0.9491 (0.9519, 0.0025)	0.9792 (0.9842, 0.0019)
	LSTM-Attention	0.9748 (0.9797, 0.0048)	0.9482 (0.9682, 0.0202)	0.9518 (0.9587, 0.0041)	0.9499 (0.9590, 0.0089)	0.9807 (0.9830, 0.0017)
	LSTM-Transformer	0.9673 (0.9806, 0.0512)	0.9169 (0.9724, 0.2159)	0.9016 (0.9560, 0.2122)	0.9092 (0.9608, 0.2140)	0.9714 (0.9837, 0.0489)
	GRU-KAN	0.9815 (0.9808, 0.0002)	0.9751 (0.9808, 0.0031)	0.9463 (0.9507, 0.0035)	0.9606 (0.9611, 0.0004)	0.9821 (0.9839, 0.0013)
	LSTM-KAN	0.9805 (0.9807, 0.0001)	0.9724 (0.9738, 0.0008)	0.9490 (0.9503, 0.0010)	0.9605 (0.9609, 0.0002)	0.9834 (0.9853, 0.0014)
18	GRU	0.9675 (0.9693, 0.0014)	0.9655 (0.9709, 0.0019)	0.9184 (0.9259, 0.0057)	0.9414 (0.9456, 0.0029)	0.9612 (0.9693, 0.0052)
	LSTM	0.9680 (0.9699, 0.0008)	0.9683 (0.9776, 0.0063)	0.9238 (0.9443, 0.0052)	0.9451 (0.9478, 0.0015)	0.9686 (0.9713, 0.0038)
	LSTM-Attention	0.9641 (0.9771, 0.0504)	0.9166 (0.9745, 0.2159)	0.8890 (0.9438, 0.2093)	0.9026 (0.9534, 0.2125)	0.9744 (0.9819, 0.0241)
	LSTM-Transformer	0.9338 (0.9804, 0.0822)	0.8264 (0.9922, 0.3563)	0.7566 (0.9460, 0.3378)	0.7877 (0.9600, 0.3437)	0.9541 (0.9832, 0.0677)
	GRU-KAN	0.9796 (0.9793, 0.0003)	0.9780 (0.9807, 0.0020)	0.9355 (0.9379, 0.0021)	0.9563 (0.9576, 0.0007)	0.9785 (0.9843, 0.0062)
	LSTM-KAN	0.9786 (0.9789, 0.0002)	0.9783 (0.9834, 0.0034)	0.9351 (0.9400, 0.0027)	0.9562 (0.9568, 0.0004)	0.9784 (0.9824, 0.0031)
21	GRU	0.9629 (0.9676, 0.0036)	0.9765 (0.9850, 0.0059)	0.8785 (0.8945, 0.0159)	0.9249 (0.9360, 0.0081)	0.9628 (0.9736, 0.0064)
	LSTM	0.9645 (0.9702, 0.0039)	0.9657 (0.9797, 0.0152)	0.8863 (0.8934, 0.0055)	0.9243 (0.9102, 0.0047)	0.9613 (0.9664, 0.0033)
	LSTM-Attention	0.9288 (0.9677, 0.0784)	0.8712 (1.0000, 0.2983)	0.7406 (0.9061, 0.3235)	0.7782 (0.9334, 0.3372)	0.9473 (0.9750, 0.0396)
	LSTM-Transformer	0.9296 (0.9720, 0.0797)	0.8052 (0.9771, 0.3479)	0.7600 (0.9335, 0.3339)	0.7812 (0.9433, 0.3396)	0.9423 (0.9737, 0.0579)
	GRU-KAN	0.9702 (0.9699, 0.0004)	0.9833 (0.9880, 0.0031)	0.8985 (0.9087, 0.0064)	0.9358 (0.9376, 0.0011)	0.9741 (0.9801, 0.0048)
	LSTM-KAN	0.9691 (0.9701, 0.0005)	0.9787 (0.9875, 0.0052)	0.9025 (0.9123, 0.0053)	0.9391 (0.9458, 0.0042)	0.9712 (0.9774, 0.0045)
24	GRU	0.9453 (0.9493, 0.0027)	0.9736 (0.9871, 0.0025)	0.8360 (0.8547, 0.0121)	0.8996 (0.9129, 0.0065)	0.9543 (0.9587, 0.0040)
	LSTM	0.9486 (0.9402, 0.0006)	0.9714 (0.9898, 0.0075)	0.8406 (0.8558, 0.0070)	0.9013 (0.9103, 0.0015)	0.9598 (0.9738, 0.0136)
	LSTM-Attention	0.9449 (0.9597, 0.0223)	0.9730 (0.9885, 0.0112)	0.8016 (0.8619, 0.0886)	0.8762 (0.9139, 0.0667)	0.9633 (0.9751, 0.0185)
	LSTM-Transformer	0.9623 (0.9717, 0.0143)	0.9509 (0.9682, 0.0166)	0.8957 (0.9319, 0.0604)	0.9213 (0.9423, 0.0346)	0.9687 (0.9743, 0.0122)
	GRU-KAN	0.9589 (0.9600, 0.0007)	0.9836 (0.9881, 0.0074)	0.8500 (0.8619, 0.0055)	0.9119 (0.9145, 0.0013)	0.9564 (0.9734, 0.0225)
	LSTM-KAN	0.9591 (0.9603, 0.0005)	0.9829 (0.9881, 0.0033)	0.8513 (0.8589, 0.0043)	0.9124 (0.9154, 0.0013)	0.9638 (0.9744, 0.0102)
27	GRU	0.9385 (0.9763, 0.0461)	0.9584 (0.9820, 0.0356)	0.7853 (0.9221, 0.1741)	0.8534 (0.9511, 0.1247)	0.9263 (0.9708, 0.0655)
	LSTM	0.9722 (0.9758, 0.0021)	0.9793 (0.9915, 0.0076)	0.9080 (0.9257, 0.0110)	0.9422 (0.9499, 0.0046)	0.9699 (0.9726, 0.0019)
	LSTM-Attention	0.9235 (0.9755, 0.0497)	0.9530 (0.9806, 0.0290)	0.7278 (0.9284, 0.1946)	0.8114 (0.9498, 0.1383)	0.9264 (0.9707, 0.0408)
	LSTM-Transformer	0.9417 (0.9726, 0.0330)	0.9512 (0.9787, 0.0238)	0.8084 (0.9233, 0.1339)	0.8673 (0.9437, 0.0913)	0.9284 (0.9689, 0.0436)
	GRU-KAN	0.9748 (0.9764, 0.0011)	0.9785 (0.9880, 0.0056)	0.9194 (0.9292, 0.0079)	0.9480 (0.9515, 0.0026)	0.9714 (0.9753, 0.0023)
	LSTM-KAN	0.9736 (0.9774, 0.0032)	0.9795 (0.9895, 0.0069)	0.9135 (0.9308, 0.0153)	0.9452 (0.9536, 0.0072)	0.9707 (0.9734, 0.0014)

Table A3Model performance on dataset *D5* across different blank intervals with a 3-month observation period.

Model	Blank Interval	Confusion Matrix				Accuracy	Precision	Recall	F1	AUC	Recall_RANK	F1_RANK	AUC_RANK
		TN	FP	FN	TP								
GRU	1	79,766	2664	162	1892	0.9665	0.4153	0.9211	0.5725	0.9665	2	3	2
	2	78,688	3742	208	1846	0.9532	0.3302	0.8982	0.4829	0.9352	9	10	14
	3	78,341	4089	247	1807	0.9486	0.3061	0.8788	0.4540	0.9214	15	14	21
	4	76,596	5834	253	1801	0.9280	0.2359	0.8768	0.3718	0.9311	16	21	16
	5	76,140	6290	342	1712	0.9215	0.2139	0.8335	0.3405	0.9109	25	25	25
	6	74,933	7497	430	1624	0.9062	0.1781	0.7907	0.2906	0.8749	33	32	36
	7	72,897	9533	501	1553	0.8812	0.1400	0.7556	0.2362	0.8734	38	44	37
	8	73,248	9182	624	1430	0.8839	0.1348	0.6962	0.2258	0.8537	46	47	45
	9	73,614	8816	855	1199	0.8854	0.1195	0.5828	0.1983	0.8003	58	58	57
	10	73,170	9260	871	1183	0.8801	0.1133	0.5759	0.1893	0.7929	59	60	62
	11	74,147	8283	993	1061	0.8902	0.1135	0.5166	0.1862	0.7958	66	64	59
	12	74,364	8066	1065	989	0.8918	0.1088	0.4796	0.1773	0.7792	68	67	66
LSTM	1	79,901	2529	178	1876	0.9679	0.4256	0.9129	0.5805	0.9614	6	2	3
	2	78,838	3592	240	1814	0.9546	0.3354	0.8827	0.4861	0.9425	13	9	10
	3	78,262	4168	294	1760	0.9472	0.2969	0.8569	0.4410	0.9225	21	15	20
	4	76,655	5775	303	1751	0.9281	0.2327	0.8525	0.3656	0.9389	22	22	13
	5	75,917	6513	406	1648	0.9182	0.2022	0.8033	0.3231	0.9093	31	28	27
	6	74,858	7572	436	1618	0.9053	0.1765	0.7897	0.2885	0.8861	34	33	34
	7	72,771	9659	514	1540	0.8796	0.1375	0.7498	0.2324	0.8725	39	46	38
	8	73,268	9162	635	1419	0.8840	0.1341	0.6908	0.2246	0.8452	48	48	49
	9	73,766	8664	768	1286	0.8882	0.1296	0.6276	0.2142	0.8230	53	49	53
	10	72,801	9629	834	1220	0.8762	0.1125	0.5940	0.1891	0.8028	56	61	54
	11	74,090	8340	978	1076	0.8897	0.1142	0.5239	0.1876	0.7870	65	62	63
	12	74,075	8355	1068	986	0.8884	0.1051	0.4776	0.1723	0.7786	69	68	67
LSTM-Attention	1	79,766	2664	166	1888	0.9665	0.4147	0.9177	0.5713	0.9561	4	4	4
	2	78,250	4180	213	1841	0.9480	0.3059	0.8968	0.4562	0.9242	10	13	18
	3	76,935	5495	276	1778	0.9317	0.2445	0.8656	0.3813	0.9169	19	19	23
	4	76,425	6005	322	1732	0.9251	0.2237	0.8423	0.3535	0.9264	23	23	17
	5	74,434	7995	361	1693	0.9010	0.1747	0.8242	0.2883	0.9099	28	34	26
	6	75,148	7282	452	1602	0.9084	0.1802	0.7795	0.2928	0.8905	35	31	32
	7	74,332	8098	597	1457	0.8971	0.1524	0.7089	0.2509	0.8711	43	41	39
	8	72,461	9969	633	1421	0.8745	0.1249	0.6928	0.2117	0.8506	47	50	48
	9	73,309	9121	751	1303	0.8832	0.1250	0.6344	0.2089	0.8287	51	52	52
	10	74,314	8116	922	1132	0.8931	0.1226	0.5516	0.2006	0.8018	61	57	55
	11	72,623	9807	938	1116	0.8728	0.1022	0.5433	0.1720	0.7838	63	69	64
	12	73,357	9073	1206	848	0.8784	0.0857	0.4138	0.1419	0.7744	72	72	68
LSTM-Transformer	1	79,376	3054	165	1889	0.9619	0.3823	0.9197	0.5401	0.9491	3	6	5
	2	77,608	4822	261	1793	0.9399	0.2715	0.8749	0.4144	0.9332	17	16	15
	3	76,687	5743	227	1827	0.9293	0.2413	0.8895	0.3796	0.9228	12	20	19
	4	75,338	7092	293	1761	0.9126	0.1993	0.8603	0.3237	0.9190	20	27	22
	5	75,139	7291	392	1662	0.9090	0.1856	0.8092	0.3019	0.9010	29	29	28
	6	74,404	8026	489	1565	0.8992	0.1633	0.7624	0.2689	0.8948	37	36	31
	7	74,135	8295	589	1465	0.8950	0.1504	0.7142	0.2485	0.8527	41	42	46
	8	72,271	10,159	613	1441	0.8726	0.1244	0.7025	0.2114	0.8649	45	51	42
	9	73,293	9137	785	1269	0.8826	0.1219	0.6178	0.2037	0.8295	55	54	51
	10	72,378	10,052	772	1282	0.8718	0.1134	0.6256	0.1920	0.7946	54	59	60
	11	75,303	7127	1153	901	0.9015	0.1128	0.4411	0.1796	0.7635	70	66	70
	12	73,136	9294	1183	871	0.8760	0.0858	0.4245	0.1427	0.7595	71	71	71
GRU-KAN	1	79,975	2455	157	1897	0.9690	0.4354	0.9221	0.5915	0.9727	1	1	1
	2	79,079	3351	189	1865	0.9580	0.3567	0.9051	0.5118	0.9485	7	7	6
	3	78,416	4014	197	1857	0.9501	0.3162	0.9036	0.4685	0.9433	8	12	9
	4	77,349	5081	245	1809	0.9369	0.2624	0.8807	0.4044	0.9418	14	17	11
	5	76,306	6124	331	1723	0.9236	0.2196	0.8393	0.3481	0.9128	24	24	24
	6	75,047	7383	423	1631	0.9075	0.1807	0.7936	0.2944	0.8977	32	30	29
	7	74,229	8201	482	1572	0.8971	0.1606	0.7648	0.2655	0.8750	36	37	35
	8	74,369	8061	574	1480	0.8977	0.1546	0.7181	0.2544	0.8553	40	39	44
	9	72,909	9521	747	1307	0.8783	0.1202	0.6339	0.2014	0.8013	52	56	56
	10	73,887	8543	852	1202	0.8887	0.1233	0.5852	0.2037	0.7945	57	55	61
	11	73,362	9068	931	1123	0.8816	0.1098	0.5448	0.1828	0.7696	62	65	69
	12	71,547	10,883	969	1085	0.8595	0.0903	0.5268	0.1542	0.7587	64	70	72
LSTM-KAN	1	79,654	2776	177	1877	0.9650	0.4033	0.9138	0.5596	0.9443	5	5	8
	2	79,162	3268	225	1829	0.9587	0.3588	0.8905	0.5115	0.9464	11	8	7
	3	78,683	3747	272	1782	0.9524	0.3220	0.8671	0.4696	0.9399	18	11	12
	4	77,421	5009	349	1705	0.9365	0.2535	0.8286	0.3882	0.8967	26	18	30
	5	75,919	6511	371	1683	0.9186	0.2064	0.8247	0.3302	0.8889	27	26	33
	6	73,901	8529	410	1644	0.8943	0.1624	0.8053	0.2703	0.8657	30	35	40
	7	74,613	7817	591	1463	0.9004	0.1575	0.7123	0.2580	0.8634	42	38	43
	8	74,529	7901	607	1447	0.8993	0.1546	0.7035	0.2535	0.8649	44	40	41
	9	74,558	7872	701	1353	0.8985	0.1464	0.6577	0.2395	0.8513	49	43	47
	10	74,576	7854	730	1324	0.8983	0.1439	0.6431	0.2352	0.8297	50	45	50
	11	74,463	7967	902	1152	0.8950	0.1263	0.5613	0.2063	0.7990	60	53	58
	12	74,187	8243	991	1063	0.8907	0.1140	0.5166	0.1868	0.7805	66	63	65

Table A4

Wilcoxon signed-rank test p -values by interval and model pair on dataset $D5$.

Metric	Interval	WG1	WG2	WG3	WG4	WL1	WL2	WL3	WL4	W5	W6
F1	1	1.83e-03**	7.66e-03*	4.25e-04***	3.15e-05***	1.00e+00	8.35e-01	6.49e-01	3.65e-03**	4.25e-04***	1.00e+00
	2	2.93e-04***	4.25e-04***	3.15e-05***	9.54e-07***	4.25e-04***	4.25e-04***	2.93e-04***	6.68e-05***	5.27e-02	9.51e-01
	3	6.81e-03*	4.25e-04***	9.54e-07***	2.86e-06***	3.65e-03*	3.54e-04***	8.39e-05***	8.39e-05***	5.94e-01	4.20e-01
	4	1.97e-04***	1.97e-04***	2.38e-05***	4.77e-06***	2.79e-03**	4.25e-04***	4.25e-04***	1.05e-04***	3.19e-03**	9.97e-01
	5	9.62e-03*	3.54e-04***	3.15e-05***	4.10e-05***	9.95e-01	9.62e-03*	2.41e-04***	4.25e-04***	2.79e-03**	9.98e-01
	6	4.48e-02	1.20e-02	1.33e-02	1.97e-04***	1.00e+00	7.74e-01	8.85e-01	1.64e-02	4.25e-04***	1.00e+00
	7	4.25e-04***	3.54e-04***	6.04e-03*	2.79e-03**	4.25e-04***	4.25e-04***	9.62e-03*	7.66e-03*	2.15e-01	7.95e-01
	8	4.25e-04***	4.25e-04***	8.39e-05***	5.25e-05***	3.54e-04***	3.54e-04***	2.93e-04***	2.93e-04***	2.42e-02	9.78e-01
	9	1.31e-01	9.98e-01	9.82e-01	9.05e-01	8.39e-05***	5.08e-04***	3.54e-04***	3.54e-04***	1.00e+00	2.93e-04***
	10	8.59e-03*	6.81e-03*	7.15e-02	9.62e-03*	5.25e-05***	2.93e-04***	4.25e-04***	2.93e-04***	1.00e+00	3.54e-04***
	11	9.05e-01	8.26e-01	9.62e-03*	4.20e-01	6.05e-04***	4.15e-03**	4.25e-04***	4.25e-04***	1.00e+00	4.25e-04***
	12	1.00e+00	1.00e+00	8.59e-03*	9.62e-03*	2.91e-02	7.66e-03*	2.41e-04***	2.93e-04***	1.00e+00	3.54e-04***
AUC	1	2.11e-03**	4.15e-03**	5.08e-04***	1.97e-04***	1.00e+00	1.00e+00	1.00e+00	1.00e+00	3.54e-04***	1.00e+00
	2	1.83e-03**	5.34e-03**	3.54e-04***	3.54e-04***	4.72e-03**	6.81e-03*	1.97e-04***	4.72e-03**	1.65e-01	8.44e-01
	3	4.25e-04***	3.54e-04***	2.93e-04***	2.93e-04***	2.93e-04***	3.54e-04***	2.41e-04***	4.25e-04***	1.23e-01	8.85e-01
	4	4.72e-03**	9.62e-02*	5.08e-04***	2.41e-04***	1.00e+00	1.00e+00	1.00e+00	1.00e+00	8.39e-05***	1.00e+00
	5	1.48e-02	8.59e-02*	1.64e-02	2.79e-03**	1.00e+00	1.00e+00	1.00e+00	1.00e+00	3.19e-03**	9.97e-01
	6	3.54e-04***	1.16e-03**	4.72e-03**	3.19e-02	1.00e+00	1.00e+00	1.00e+00	1.00e+00	2.93e-04***	1.00e+00
	7	1.84e-01	7.66e-03*	8.59e-03*	2.93e-04***	1.00e+00	1.00e+00	1.00e+00	7.66e-03*	2.79e-03**	9.98e-01
	8	1.84e-01	4.15e-03**	7.66e-03*	1.00e+00	4.72e-03**	3.54e-04***	2.79e-03**	3.24e-01	9.96e-01	4.72e-03**
	9	4.78e-01	1.00e+00	1.00e+00	1.00e+00	6.68e-05***	2.93e-04***	3.54e-04***	3.54e-04***	1.00e+00	5.25e-05***
	10	1.74e-01	1.00e+00	1.00e+00	1.00e+00	1.05e-04***	2.93e-04***	2.93e-04***	2.93e-04***	1.00e+00	1.05e-04***
	11	1.00e+00	1.00e+00	1.00e+00	4.72e-03**	7.66e-03**	2.11e-03**	4.25e-04***	1.05e-04***	1.00e+00	1.31e-04***
	12	1.00e+00	1.00e+00	1.00e+00	1.00e+00	2.91e-02	4.87e-02	5.34e-03**	4.25e-04***	1.00e+00	2.93e-04***
Recall	1	4.78e-01	7.66e-03*	1.07e-02	4.64e-01	9.23e-01	1.64e-02	8.35e-01	6.63e-01	8.59e-03*	9.92e-01
	2	8.59e-03*	3.54e-04***	6.81e-03*	3.54e-04***	9.47e-01	6.04e-03*	8.26e-01	4.25e-04***	4.15e-03**	9.96e-01
	3	3.54e-04***	1.61e-04***	2.41e-04***	3.65e-03**	1.00e+00	9.90e-04**	2.20e-02	5.94e-01	3.54e-04***	1.00e+00
	4	4.64e-01	2.93e-04***	2.41e-04***	4.25e-04***	1.00e+00	1.00e+00	9.97e-01	7.95e-01	2.41e-04***	1.00e+00
	5	9.62e-03*	1.97e-04***	5.08e-04***	3.54e-04***	9.68e-01	3.54e-04***	1.64e-02	5.08e-04***	3.65e-03**	9.97e-01
	6	8.84e-02	1.07e-02	9.90e-04**	2.93e-04***	4.15e-03**	4.25e-04***	3.54e-04***	2.93e-04***	9.96e-01	4.15e-03**
	7	6.04e-03*	4.25e-04***	1.34e-05***	4.10e-05***	1.00e+00	1.00e+00	1.33e-02	7.51e-01	1.97e-04***	1.00e+00
	8	2.41e-04***	3.54e-04***	2.93e-04***	5.08e-04***	9.62e-03*	4.15e-03**	9.62e-03*	3.11e-01	4.15e-03**	9.96e-01
	9	1.81e-05***	7.27e-01	9.65e-01	5.08e-04***	2.38e-05***	4.25e-04***	4.25e-04***	3.54e-04***	1.00e+00	3.54e-04***
	10	6.04e-03*	9.99e-01	2.41e-04***	9.89e-01	3.15e-05***	2.93e-04***	1.34e-05***	5.08e-04***	1.00e+00	1.61e-04***
	11	2.93e-04***	5.08e-04***	4.35e-01	9.54e-06***	2.41e-04***	3.54e-04***	5.08e-04***	9.54e-07***	9.99e-01	9.90e-04**
	12	1.31e-04***	1.05e-04***	6.68e-06***	9.54e-07***	3.54e-04***	3.54e-04***	9.54e-07***	6.68e-06***	8.59e-03*	9.92e-01

* Significant at the unadjusted level α .

** Significant after Benjamini-Hochberg (BH) FDR adjustment.

*** Significant after Benjamini-Yekutieli (BY) FDR adjustment.

Table A5
Results of model performance based on 20 trials using different sample sizes.

Sample size	Model	Accuracy (Best, Std)	Precision (Best, Std)	Recall (Best, Std)	F1 (Best, Std)	AUC (Best, Std)
500,000	GRU	0.8769 (0.9825, 0.1102)	0.5911 (1.0000, 0.4951)	0.5157 (0.9484, 0.4485)	0.5453 (0.9644, 0.4643)	0.9056 (0.9775, 0.0766)
	LSTM	0.9577 (0.9603, 0.0024)	0.9594 (0.9796, 0.0184)	0.8536 (0.8765, 0.0148)	0.9034 (0.9169, 0.0081)	0.9637 (0.9721, 0.0066)
	LSTM-Attention	0.8019 (0.9833, 0.0912)	0.4742 (1.0000, 0.4877)	0.2121 (0.9541, 0.3721)	0.2931 (0.9662, 0.3878)	0.8516 (0.9773, 0.1015)
	LSTM-Transformer	0.9001 (0.9544, 0.0811)	0.9858 (1.0000, 0.0105)	0.6119 (0.8378, 0.3314)	0.7551 (0.9018, 0.3551)	0.9001 (0.9142, 0.0047)
	GRU-KAN	0.9707 (0.9851, 0.0137)	0.9874 (1.0000, 0.0059)	0.8946 (0.9554, 0.0581)	0.9376 (0.9698, 0.0326)	0.9681 (0.9881, 0.0225)
	LSTM-KAN	0.9638 (0.9851, 0.0264)	0.9831 (0.9939, 0.0071)	0.8705 (0.9604, 0.1098)	0.9190 (0.9700, 0.0738)	0.9744 (0.9900, 0.0175)
1,000,000	GRU	0.8885 (0.9827, 0.1160)	0.5857 (0.9776, 0.4907)	0.5679 (0.9536, 0.4758)	0.5767 (0.9651, 0.4831)	0.9094 (0.9783, 0.0882)
	LSTM	0.9671 (0.9825, 0.0126)	0.9612 (0.9820, 0.0114)	0.8954 (0.9570, 0.0528)	0.9271 (0.9568, 0.0286)	0.9697 (0.9758, 0.0046)
	LSTM-Attention	0.8465 (0.9714, 0.1025)	0.4694 (0.9769, 0.4818)	0.4144 (0.9458, 0.4417)	0.4343 (0.9427, 0.4524)	0.9046 (0.9860, 0.0622)
	LSTM-Transformer	0.9078 (0.9788, 0.0843)	0.8876 (1.0000, 0.3036)	0.6416 (0.9331, 0.3427)	0.7448 (0.9565, 0.3658)	0.9171 (0.9839, 0.0414)
	GRU-KAN	0.9767 (0.9854, 0.0097)	0.9803 (0.9912, 0.0116)	0.9257 (0.9541, 0.0358)	0.9519 (0.9703, 0.0210)	0.9757 (0.9874, 0.0173)
	LSTM-KAN	0.9811 (0.9854, 0.0053)	0.9773 (0.9870, 0.0069)	0.9462 (0.9582, 0.0199)	0.9614 (0.9704, 0.0114)	0.9845 (0.9898, 0.0075)
1,500,000	GRU	0.9456 (0.9625, 0.0149)	0.8783 (0.9805, 0.3004)	0.8329 (0.9475, 0.2853)	0.8550 (0.9602, 0.2925)	0.9376 (0.9697, 0.0308)
	LSTM	0.9721 (0.9840, 0.0014)	0.9622 (0.9706, 0.0053)	0.9349 (0.9382, 0.0027)	0.9484 (0.9531, 0.0089)	0.9767 (0.9788, 0.0014)
	LSTM-Attention	0.9143 (0.9618, 0.0571)	0.9340 (1.0000, 0.2203)	0.6706 (0.8706, 0.2365)	0.7807 (0.9193, 0.2326)	0.9153 (0.9284, 0.0154)
	LSTM-Transformer	0.9531 (0.9804, 0.0696)	0.9510 (0.9782, 0.0174)	0.8324 (0.8959, 0.1170)	0.8878 (0.9217, 0.0810)	0.9493 (0.9746, 0.0557)
	GRU-KAN	0.9798 (0.9841, 0.0124)	0.9727 (0.9873, 0.0425)	0.9482 (0.9616, 0.0091)	0.9597 (0.9678, 0.0216)	0.9835 (0.9868, 0.0033)
	LSTM-KAN	0.9823 (0.9835, 0.0010)	0.9817 (0.9848, 0.0035)	0.9468 (0.9577, 0.0062)	0.9639 (0.9665, 0.0021)	0.9864 (0.9882, 0.0011)
2,000,000	GRU	0.9406 (0.9659, 0.0245)	0.8948 (0.8972, 0.0135)	0.8568 (0.8844, 0.0377)	0.8754 (0.8897, 0.0078)	0.9461 (0.9708, 0.0326)
	LSTM	0.9718 (0.9834, 0.0110)	0.9613 (0.9790, 0.0152)	0.9456 (0.9511, 0.0087)	0.9534 (0.9563, 0.0019)	0.9858 (0.9885, 0.0019)
	LSTM-Attention	0.9437 (0.9659, 0.0424)	0.9702 (0.9890, 0.0083)	0.7999 (0.8883, 0.1748)	0.8769 (0.9287, 0.1595)	0.9487 (0.9747, 0.0324)
	LSTM-Transformer	0.9449 (0.9666, 0.0181)	0.9675 (0.9905, 0.0255)	0.8075 (0.8935, 0.0741)	0.8782 (0.9300, 0.0444)	0.9320 (0.9706, 0.0345)
	GRU-KAN	0.9823 (0.9830, 0.0007)	0.9747 (0.9787, 0.0044)	0.9539 (0.9596, 0.0024)	0.9642 (0.9656, 0.0013)	0.9853 (0.9881, 0.0023)
	LSTM-KAN	0.9821 (0.9837, 0.0010)	0.9732 (0.9807, 0.0058)	0.9547 (0.9596, 0.0032)	0.9638 (0.9670, 0.0018)	0.9865 (0.9894, 0.0013)
3,000,000	GRU	0.9565 (0.9669, 0.0505)	0.9246 (0.9612, 0.0180)	0.8797 (0.8908, 0.0083)	0.9016 (0.9207, 0.0643)	0.9334 (0.9663, 0.0459)
	LSTM	0.9764 (0.9828, 0.0012)	0.9684 (0.9785, 0.0058)	0.9504 (0.9569, 0.0018)	0.9593 (0.9651, 0.0023)	0.9847 (0.9865, 0.0020)
	LSTM-Attention	0.9616 (0.9791, 0.0499)	0.9182 (0.9706, 0.1619)	0.8986 (0.9594, 0.2117)	0.9083 (0.9575, 0.1915)	0.9773 (0.9856, 0.0266)
	LSTM-Transformer	0.9396 (0.9654, 0.0496)	0.9126 (0.9868, 0.2215)	0.7893 (0.9014, 0.2077)	0.8440 (0.9277, 0.2101)	0.9336 (0.9647, 0.0402)
	GRU-KAN	0.9838 (0.9847, 0.0017)	0.9765 (0.9814, 0.0100)	0.9524 (0.9581, 0.0041)	0.9643 (0.9651, 0.0032)	0.9859 (0.9872, 0.0022)
	LSTM-KAN	0.9807 (0.9829, 0.0026)	0.9714 (0.9814, 0.0138)	0.9511 (0.9581, 0.0045)	0.9611 (0.9652, 0.0049)	0.9839 (0.9861, 0.0020)
5,000,000	GRU	0.9793 (0.9817, 0.0023)	0.9585 (0.9841, 0.0095)	0.9211 (0.9517, 0.0109)	0.9394 (0.9656, 0.0048)	0.9736 (0.9785, 0.0026)
	LSTM	0.9814 (0.9828, 0.0012)	0.9716 (0.9785, 0.0058)	0.9534 (0.9569, 0.0018)	0.9624 (0.9664, 0.0025)	0.9847 (0.9865, 0.0020)
	LSTM-Attention	0.9776 (0.9812, 0.0027)	0.9548 (0.9697, 0.0116)	0.9157 (0.9619, 0.2040)	0.9348 (0.9558, 0.0059)	0.9754 (0.9773, 0.0018)
	LSTM-Transformer	0.9808 (0.9836, 0.0043)	0.9753 (0.9799, 0.0031)	0.9368 (0.9497, 0.0425)	0.9557 (0.9594, 0.0155)	0.9829 (0.9846, 0.0012)
	GRU-KAN	0.9854 (0.9869, 0.0015)	0.9790 (0.9808, 0.0009)	0.9588 (0.9658, 0.0052)	0.9688 (0.9705, 0.0031)	0.9858 (0.9885, 0.0019)
	LSTM-KAN	0.9822 (0.9825, 0.0005)	0.9762 (0.9833, 0.0062)	0.9491 (0.9525, 0.0027)	0.9625 (0.9645, 0.0012)	0.9837 (0.9856, 0.0016)

Table A6

Results of multi-year experiment performance based on 20 trials using different cohorts of training and test sets (feature window = 15, blank interval = 3, observation period = 3).

Training→Test	Model	Accuracy (Best, Std)	Precision (Best, Std)	Recall (Best, Std)	F1 (Best, Std)	AUC (Best, Std)
2018→2020	GRU	0.9914 (0.9921, 0.0008)	0.2844 (0.3066, 0.0236)	0.7836 (0.8024, 0.0249)	0.4173 (0.4277, 0.0218)	0.8474 (0.8789, 0.0153)
	LSTM	0.9916 (0.9924, 0.0009)	0.2934 (0.3177, 0.0296)	0.8041 (0.8254, 0.0439)	0.4300 (0.4429, 0.0311)	0.8492 (0.8827, 0.0176)
	LSTM-Attention	0.9844 (0.9852, 0.0009)	0.1714 (0.2028, 0.0365)	0.7743 (0.7990, 0.0538)	0.2806 (0.2815, 0.0196)	0.8026 (0.8644, 0.0465)
	LSTM-Transformer	0.9806 (0.9814, 0.0011)	0.1381 (0.2469, 0.1114)	0.7530 (0.7785, 0.0363)	0.2334 (0.2312, 0.0121)	0.7508 (0.7868, 0.0543)
	GRU-KAN	0.9937 (0.9942, 0.0005)	0.3726 (0.3925, 0.0207)	0.8654 (0.8782, 0.0195)	0.5209 (0.5263, 0.0067)	0.8655 (0.8761, 0.0107)
	LSTM-KAN	0.9939 (0.9944, 0.0005)	0.3746 (0.3972, 0.0233)	0.8322 (0.8433, 0.0211)	0.5167 (0.5239, 0.0092)	0.8579 (0.8703, 0.0114)
2018→2021	GRU	0.9919 (0.9924, 0.0007)	0.2509 (0.2687, 0.0259)	0.7272 (0.7490, 0.0439)	0.3730 (0.3794, 0.0280)	0.8338 (0.8679, 0.0165)
	LSTM	0.9922 (0.9927, 0.0007)	0.2597 (0.2781, 0.0248)	0.7406 (0.7593, 0.0384)	0.3845 (0.3916, 0.0258)	0.8162 (0.8543, 0.0240)
	LSTM-Attention	0.9845 (0.9851, 0.0051)	0.1442 (0.2522, 0.1113)	0.6093 (0.5674, 0.0452)	0.2332 (0.1850, 0.0244)	0.7914 (0.8276, 0.0093)
	LSTM-Transformer	0.9808 (0.9814, 0.0009)	0.1079 (0.2245, 0.1067)	0.6619 (0.6879, 0.0493)	0.1855 (0.1801, 0.0125)	0.7678 (0.8376, 0.0592)
	GRU-KAN	0.9934 (0.9937, 0.0004)	0.3105 (0.3251, 0.0158)	0.8192 (0.8276, 0.0165)	0.4503 (0.4537, 0.0135)	0.8420 (0.8548, 0.0064)
	LSTM-KAN	0.9932 (0.9936, 0.0003)	0.3036 (0.3186, 0.0138)	0.8075 (0.8184, 0.0166)	0.4412 (0.4455, 0.0110)	0.8315 (0.8720, 0.0265)
2018→2022	GRU	0.9545 (0.9562, 0.0016)	0.0681 (0.0743, 0.0062)	0.4418 (0.4669, 0.0289)	0.1181 (0.1250, 0.0089)	0.6356 (0.7125, 0.0553)
	LSTM	0.9579 (0.9597, 0.0017)	0.0719 (0.0802, 0.0081)	0.4292 (0.4627, 0.0352)	0.1232 (0.1335, 0.0116)	0.6365 (0.7043, 0.0485)
	LSTM-Attention	0.9496 (0.9515, 0.0019)	0.0579 (0.0648, 0.0075)	0.4134 (0.4502, 0.0432)	0.1016 (0.1080, 0.0104)	0.6467 (0.6970, 0.1308)
	LSTM-Transformer	0.9442 (0.9462, 0.0022)	0.0491 (0.0558, 0.0065)	0.3861 (0.4281, 0.0394)	0.0871 (0.0935, 0.0087)	0.6357 (0.7015, 0.1151)
	GRU-KAN	0.9701 (0.9709, 0.0007)	0.1371 (0.1429, 0.0044)	0.6327 (0.6443, 0.0095)	0.2253 (0.2285, 0.0037)	0.7624 (0.7781, 0.0245)
	LSTM-KAN	0.9711 (0.9721, 0.0007)	0.1410 (0.1478, 0.0051)	0.6264 (0.6411, 0.0108)	0.2302 (0.2353, 0.0048)	0.7760 (0.7947, 0.0199)
2019→2021	GRU	0.9912 (0.9920, 0.0006)	0.2207 (0.2457, 0.0184)	0.6536 (0.6810, 0.0194)	0.3302 (0.3483, 0.0161)	0.7942 (0.8156, 0.0203)
	LSTM	0.9913 (0.9925, 0.0007)	0.2246 (0.2466, 0.0203)	0.6611 (0.6814, 0.0168)	0.3352 (0.3487, 0.0186)	0.8073 (0.8679, 0.0319)
	LSTM-Attention	0.9908 (0.9916, 0.0008)	0.2121 (0.2344, 0.0204)	0.6561 (0.6773, 0.0186)	0.3206 (0.3354, 0.0192)	0.7855 (0.8159, 0.0393)
	LSTM-Transformer	0.9902 (0.9908, 0.0008)	0.1976 (0.2149, 0.0183)	0.6452 (0.6676, 0.0221)	0.3025 (0.3124, 0.0181)	0.7473 (0.8061, 0.0774)
	GRU-KAN	0.9927 (0.9931, 0.0004)	0.2700 (0.2842, 0.0136)	0.7029 (0.713, 0.0144)	0.3902 (0.3936, 0.0106)	0.8184 (0.8248, 0.0086)
	LSTM-KAN	0.9926 (0.9932, 0.0003)	0.2626 (0.2755, 0.0121)	0.6812 (0.6929, 0.0109)	0.3790 (0.3811, 0.0084)	0.8075 (0.8163, 0.0077)
2019→2022	GRU	0.9732 (0.9743, 0.0016)	0.1333 (0.1435, 0.0168)	0.5257 (0.5477, 0.0501)	0.2127 (0.2143, 0.0205)	0.7237 (0.7641, 0.0532)
	LSTM	0.9736 (0.9747, 0.0016)	0.1397 (0.1510, 0.0154)	0.5498 (0.5764, 0.0408)	0.2227 (0.2262, 0.0177)	0.7365 (0.7792, 0.0448)
	LSTM-Attention	0.9724 (0.9745, 0.0021)	0.1280 (0.1439, 0.0185)	0.5163 (0.5444, 0.0478)	0.2051 (0.2135, 0.0213)	0.7053 (0.7423, 0.0832)
	LSTM-Transformer	0.9701 (0.9722, 0.0019)	0.1133 (0.1285, 0.0204)	0.4921 (0.5257, 0.0652)	0.1842 (0.1924, 0.0257)	0.6978 (0.7378, 0.1049)
	GRU-KAN	0.9797 (0.9806, 0.0008)	0.1900 (0.2019, 0.0110)	0.5960 (0.6128, 0.0211)	0.2882 (0.2896, 0.0089)	0.7634 (0.7831, 0.0153)
	LSTM-KAN	0.9788 (0.9797, 0.0007)	0.1824 (0.1929, 0.0084)	0.595 (0.6086, 0.0121)	0.2792 (0.2888, 0.0091)	0.7535 (0.7673, 0.0162)
2020→2022	GRU	0.9816 (0.9834, 0.0018)	0.2232 (0.2488, 0.0232)	0.6716 (0.6966, 0.0205)	0.3351 (0.3525, 0.0217)	0.7525 (0.7644, 0.0252)
	LSTM	0.9825 (0.9841, 0.0018)	0.2363 (0.2624, 0.0242)	0.6905 (0.7177, 0.0178)	0.3521 (0.3681, 0.0213)	0.7798 (0.8084, 0.0219)
	LSTM-Attention	0.9815 (0.9835, 0.0023)	0.1981 (0.2268, 0.0448)	0.5519 (0.5769, 0.0971)	0.2915 (0.3095, 0.0553)	0.7603 (0.7995, 0.0193)
	LSTM-Transformer	0.9689 (0.9710, 0.0026)	0.1119 (0.1256, 0.0291)	0.5068 (0.5373, 0.1171)	0.1833 (0.1874, 0.0407)	0.7464 (0.7874, 0.0054)
	GRU-KAN	0.9828 (0.9838, 0.0009)	0.2413 (0.2569, 0.0161)	0.6957 (0.7135, 0.0251)	0.3583 (0.3616, 0.0139)	0.7805 (0.8026, 0.0045)
	LSTM-KAN	0.9832 (0.9842, 0.0009)	0.2485 (0.2651, 0.0152)	0.7135 (0.7324, 0.0205)	0.3687 (0.3731, 0.0117)	0.7912 (0.8093, 0.0049)

Table A7

Results of cross-quarter experiment performance based on 20 trials using different cohorts of training and test sets (feature window = 15, blank interval = 3, observation period = 3).

Training→Test	Model	Accuracy (Best, Std)	Precision (Best, Std)	Recall (Best, Std)	F1 (Best, Std)	AUC (Best, Std)
18Q1→18Q2	GRU	0.9648 (0.9666, 0.0019)	0.1654 (0.1818, 0.0188)	0.6367 (0.6857, 0.0619)	0.2626 (0.2715, 0.0219)	0.8506 (0.8662, 0.0229)
	LSTM	0.9687 (0.9709, 0.0021)	0.1871 (0.2094, 0.0236)	0.6531 (0.7061, 0.0491)	0.2909 (0.3071, 0.0277)	0.8385 (0.8785, 0.0328)
	LSTM-Attention	0.9085 (0.9103, 0.0019)	0.0705 (0.0773, 0.0079)	0.6816 (0.7429, 0.0409)	0.1278 (0.1341, 0.0116)	0.8320 (0.8626, 0.0291)
	LSTM-Transformer	0.9091 (0.9121, 0.0027)	0.0712 (0.0845, 0.0116)	0.6827 (0.8082, 0.0429)	0.1290 (0.1495, 0.0191)	0.8073 (0.8523, 0.0371)
	GRU-KAN	0.9888 (0.9898, 0.0009)	0.4481 (0.4882, 0.0345)	0.6163 (0.6776, 0.0335)	0.5189 (0.5516, 0.0331)	0.8604 (0.8762, 0.0241)
	LSTM-KAN	0.9889 (0.9898, 0.0011)	0.4529 (0.4880, 0.0382)	0.6857 (0.6612, 0.0289)	0.5192 (0.5456, 0.0367)	0.8622 (0.8882, 0.0294)
18Q1→18Q3	GRU	0.8748 (0.8771, 0.0025)	0.1014 (0.1095, 0.0092)	0.6086 (0.6557, 0.0265)	0.1738 (0.1812, 0.0127)	0.7748 (0.8099, 0.0320)
	LSTM	0.9292 (0.9317, 0.0027)	0.1741 (0.1894, 0.0171)	0.6063 (0.6569, 0.0272)	0.2706 (0.2841, 0.0217)	0.7824 (0.8247, 0.0342)
	LSTM-Attention	0.9389 (0.9413, 0.0027)	0.1990 (0.2143, 0.0191)	0.6018 (0.6425, 0.0283)	0.2991 (0.3055, 0.0217)	0.7867 (0.8427, 0.0442)
	LSTM-Transformer	0.9591 (0.9613, 0.0023)	0.2819 (0.3064, 0.0222)	0.5747 (0.6218, 0.0388)	0.3783 (0.3946, 0.0202)	0.7925 (0.8275, 0.0321)
	GRU-KAN	0.9854 (0.9865, 0.0012)	0.6774 (0.7075, 0.0296)	0.6176 (0.6403, 0.0272)	0.6462 (0.6563, 0.0205)	0.8217 (0.8473, 0.0262)
	LSTM-KAN	0.9641 (0.9655, 0.0014)	0.3197 (0.3367, 0.0176)	0.5814 (0.6109, 0.0328)	0.4125 (0.4181, 0.0156)	0.8016 (0.8272, 0.0266)
18Q1→18Q4	GRU	0.7592 (0.7628, 0.0031)	0.0875 (0.0945, 0.0063)	0.4559 (0.4918, 0.0309)	0.1468 (0.1491, 0.0055)	0.6535 (0.6791, 0.0273)
	LSTM	0.7126 (0.7168, 0.0035)	0.0811 (0.0886, 0.0059)	0.5157 (0.5635, 0.0252)	0.1402 (0.1472, 0.0072)	0.6169 (0.6569, 0.0354)
	LSTM-Attention	0.9223 (0.9268, 0.0041)	0.2121 (0.2534, 0.0374)	0.2616 (0.3139, 0.0332)	0.2343 (0.2645, 0.0341)	0.6381 (0.6631, 0.0285)
	LSTM-Transformer	0.9315 (0.9365, 0.0047)	0.2474 (0.3038, 0.0517)	0.2481 (0.3079, 0.0443)	0.2478 (0.2899, 0.0444)	0.6263 (0.6503, 0.0249)
	GRU-KAN	0.9311 (0.9328, 0.0021)	0.3213 (0.3337, 0.0161)	0.4649 (0.4813, 0.0219)	0.3800 (0.3882, 0.0172)	0.6736 (0.7012, 0.0289)
	LSTM-KAN	0.9460 (0.9474, 0.0015)	0.3568 (0.3826, 0.0254)	0.2347 (0.2556, 0.0382)	0.2831 (0.2905, 0.0129)	0.6670 (0.6946, 0.0295)
18Q1→19Q1	GRU	0.8617 (0.8661, 0.0038)	0.1105 (0.1295, 0.0153)	0.3019 (0.3552, 0.0276)	0.1617 (0.1849, 0.0198)	0.6148 (0.6498, 0.0314)
	LSTM	0.8770 (0.8808, 0.0035)	0.1208 (0.1376, 0.0158)	0.2841 (0.3228, 0.0153)	0.1695 (0.1881, 0.0195)	0.5938 (0.6368, 0.0262)
	LSTM-Attention	0.8153 (0.8192, 0.0037)	0.0889 (0.0997, 0.0101)	0.3442 (0.3853, 0.0223)	0.1413 (0.1535, 0.0136)	0.6117 (0.6467, 0.0274)
	LSTM-Transformer	0.8894 (0.8933, 0.0042)	0.0590 (0.0846, 0.0243)	0.1006 (0.1442, 0.0057)	0.0744 (0.1017, 0.0281)	0.5843 (0.6199, 0.0224)
	GRU-KAN	0.8888 (0.8909, 0.0023)	0.1669 (0.1774, 0.0096)	0.3799 (0.4042, 0.0154)	0.2319 (0.2416, 0.0107)	0.6313 (0.6556, 0.0221)
	LSTM-KAN	0.8035 (0.8052, 0.0019)	0.0848 (0.0895, 0.0051)	0.3523 (0.3718, 0.0115)	0.1367 (0.1393, 0.0063)	0.6045 (0.6387, 0.0285)
18Q2→18Q3	GRU	0.9378 (0.9394, 0.0021)	0.2022 (0.2121, 0.0137)	0.6357 (0.6629, 0.0183)	0.3068 (0.3165, 0.0185)	0.8253 (0.8609, 0.0294)
	LSTM	0.9134 (0.9157, 0.0022)	0.1521 (0.1651, 0.0102)	0.6561 (0.7134, 0.0126)	0.2469 (0.2632, 0.0138)	0.8181 (0.8562, 0.0324)
	LSTM-Attention	0.9478 (0.9502, 0.0026)	0.2349 (0.2564, 0.0215)	0.6244 (0.6851, 0.0283)	0.3414 (0.3682, 0.0287)	0.8389 (0.8775, 0.0333)
	LSTM-Transformer	0.9119 (0.9156, 0.0033)	0.1454 (0.1631, 0.0167)	0.6290 (0.7014, 0.0315)	0.2362 (0.2597, 0.0244)	0.8057 (0.8309, 0.0191)
	GRU-KAN	0.9497 (0.9513, 0.0015)	0.2498 (0.2641, 0.0137)	0.6606 (0.6991, 0.0213)	0.3625 (0.3784, 0.0177)	0.8392 (0.8648, 0.0261)
	LSTM-KAN	0.9474 (0.9489, 0.0016)	0.2306 (0.2432, 0.0143)	0.6109 (0.6448, 0.0342)	0.3348 (0.3482, 0.0183)	0.8388 (0.8639, 0.0294)
18Q2→18Q4	GRU	0.8944 (0.8982, 0.0035)	0.1785 (0.1991, 0.0178)	0.3677 (0.4101, 0.0228)	0.2404 (0.2631, 0.0212)	0.7281 (0.7732, 0.0359)
	LSTM	0.8652 (0.8693, 0.0035)	0.1541 (0.1698, 0.0134)	0.4380 (0.4826, 0.0243)	0.2281 (0.2463, 0.0173)	0.7133 (0.7516, 0.0287)
	LSTM-Attention	0.9101 (0.9132, 0.0033)	0.2135 (0.2336, 0.0204)	0.3647 (0.3991, 0.0278)	0.2693 (0.2898, 0.0233)	0.7349 (0.7794, 0.0385)
	LSTM-Transformer	0.9048 (0.9093, 0.0049)	0.1789 (0.2082, 0.0315)	0.3049 (0.3558, 0.0306)	0.2255 (0.2578, 0.0373)	0.7283 (0.7563, 0.0254)
	GRU-KAN	0.9067 (0.9089, 0.0021)	0.2404 (0.2509, 0.0106)	0.4873 (0.5067, 0.0194)	0.3220 (0.3307, 0.0113)	0.7397 (0.7699, 0.0276)
	LSTM-KAN	0.8893 (0.8917, 0.0023)	0.1957 (0.2054, 0.0099)	0.4619 (0.4828, 0.0182)	0.2749 (0.2833, 0.0114)	0.7255 (0.7501, 0.0224)
18Q2→19Q1	GRU	0.9391 (0.9433, 0.0041)	0.2798 (0.3391, 0.0542)	0.2403 (0.2987, 0.0214)	0.2585 (0.3127, 0.0488)	0.6459 (0.7195, 0.0495)
	LSTM	0.9073 (0.9097, 0.0034)	0.1632 (0.1797, 0.0217)	0.2662 (0.2927, 0.0185)	0.2023 (0.2178, 0.0243)	0.6230 (0.7081, 0.0562)
	LSTM-Attention	0.9221 (0.9242, 0.0034)	0.1905 (0.2123, 0.0298)	0.2354 (0.2643, 0.0224)	0.2106 (0.2306, 0.0306)	0.6316 (0.7167, 0.0551)
	LSTM-Transformer	0.9296 (0.9331, 0.0041)	0.2185 (0.2535, 0.0382)	0.2305 (0.2646, 0.0234)	0.2243 (0.2541, 0.0351)	0.6441 (0.7101, 0.0462)
	GRU-KAN	0.9350 (0.9371, 0.0018)	0.2762 (0.2971, 0.0177)	0.2906 (0.3101, 0.0175)	0.2832 (0.2985, 0.0145)	0.6835 (0.7495, 0.0413)
	LSTM-KAN	0.9362 (0.9386, 0.0021)	0.2850 (0.3119, 0.0239)	0.2938 (0.3231, 0.0187)	0.2894 (0.3124, 0.0224)	0.6873 (0.7509, 0.0434)
18Q2→19Q2	GRU	0.9305 (0.9327, 0.0022)	0.3102 (0.3226, 0.0124)	0.7152 (0.7425, 0.0273)	0.4327 (0.4448, 0.0147)	0.8555 (0.8706, 0.0176)
	LSTM	0.9260 (0.9281, 0.0021)	0.2955 (0.3058, 0.0104)	0.7217 (0.7431, 0.0212)	0.4193 (0.4283, 0.0117)	0.8439 (0.8669, 0.0327)
	LSTM-Attention	0.9366 (0.9387, 0.0022)	0.3325 (0.3447, 0.0128)	0.7048 (0.7281, 0.0232)	0.4519 (0.4633, 0.0142)	0.8666 (0.8933, 0.0248)
	LSTM-Transformer	0.9328 (0.9353, 0.0025)	0.3181 (0.3318, 0.0141)	0.7101 (0.7373, 0.0273)	0.4393 (0.4527, 0.0162)	0.8520 (0.8774, 0.0207)
	GRU-KAN	0.9410 (0.9427, 0.0016)	0.3553 (0.3671, 0.0103)	0.7282 (0.7535, 0.0206)	0.4776 (0.4887, 0.0112)	0.8667 (0.8928, 0.0248)
	LSTM-KAN	0.9352 (0.9371, 0.0017)	0.3288 (0.3401, 0.0106)	0.7191 (0.7446, 0.0244)	0.4512 (0.4621, 0.0124)	0.8655 (0.8907, 0.0276)

Data availability

Availability of data and material: The dataset analyzed during the current study is available in the Freddie Mac repository, <https://www.freddiemac.com/research/datasets/sf-loanlevel-dataset>.

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